RITE-HITE®
ALWAYS LOOKING AHEAD

SHR-5000™ Dok-Lok®

Shadow Hook Vehicle Restraint

Owner's Manual

This manual covers units: built after serial numbers: 4487000001 and up



9in Carriage



12in Carriage

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Model: **SHR-5000**

Language: **English**

Publication: **AMEN00096 2018-05-30**

NOTICE TO USER

Thank you for purchasing a SHR-5000™ Dok-Lok®.

SHR-5000 Dok-Lok vehicle restraint is an electromechanical restraint device that, when properly installed and operated, retains a secure connection between truck and dock. Signal lights, warning horn and signs provide instructions to truck driver and Dok-Lok vehicle restraint operator that a safe condition exists. The Dok-Lok vehicle restraint is operated by pressing push buttons on an interior control panel.

The English version of this manual shall prevail over any error in, or conflicting interpretation of, any translations.

Read and understand contents of this manual prior to installation or operation of this equipment.

Rite-Hite reserves the right to substitute and/or modify parts and drawings (electrical & architectural). If separate prints are included with unit, they supersede manual.

Contact Rite-Hite Customer Support for installation help not covered in this manual: (U.S.) 800-456-0600.

A Planned Maintenance Program (P.M.P.), customized to your specific operation is available and recommended. For a P.M.P., contact your local Rite-Hite representative or Rite-Hite technical support at (U.S.) 414-355-2600.

The Rite-Hite® products in this manual are covered by one or more of the following U.S. patents: 5882167, 6065172, 6070283, 6085375, 6092970, 6106212, 6116839, 6190109, 6276016, 6311352, 6318947, 6322310, 6360394, 6368043, 6431819, 6488464, 6524053, 6726432, 6773221, 6832403, 6880301, 7032267, 7062814, 7213285, 7216391, 7363670, 7380305, 7503089, 7533431, 7546655, 7584517, 7681271, 7823239, 7841823, 7877831, 7914042, 8006811, 8065770, 8141189, 8191194, 8286757, 8287223, 8303235, 8307956, 8443474, 8464384, 8464846, 8465245, 8497761, 8499897, 8544130, 8547234, 8590087, 8590673, 8616826, 8657551, 8662535, 8678736, 8690087, 8905198, 9010501, 9096170, 9096397, 9126775, 9139384, 9145273, 9150367, 9174811, 9227799, 9230419 and may be covered by additional pending U.S. and foreign patent applications.

Rite-Hite®, ThinMan™, Safe-T-Lip®, Hydrachek®, Wheel-Lok™, Dok-Lok®, Dual-Dok®, Safe-T-Strut™, Dok-Commander®, Jumbo™, Hydra-Rite™, Safe-T-Gate®, Rite-Vu™ Hazard Recognition and Communication System, and Smooth Transition Dok System™, are trademarks of Rite-Hite®.

FCC Compliance

NOTE: This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

NOTE: Changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions:

1. This device may not cause harmful interference.
2. This device must accept any interference received, including interference that may cause undesirable operation.

NOTICE TO USER

Owner Responsibility

1. The owner should recognize the inherent danger of the interface between dock and transport vehicle. The owner should, therefore, train and instruct operators in the safe use of dock equipment in accordance with the information provided below. The manufacturer shall publish, provide to the initial purchaser, and make the following information readily available to owners:
   - Installation instructions
   - Recommended initial and periodic inspections procedures
   - Maintenance procedures
   - Operating instructions
   - Descriptions or specifications for replaceable or repairable parts
   - Tables identifying the grade (slope) for all variations of length or configuration of the dock equipment, and
   - Information identifying the maximum uncontrolled drop encountered upon sudden removal of support while within the working range of the equipment.

   It shall be the responsibility of the owner to verify that the material listed in this section has been received and that it is made available for the instruction and training of personnel entrusted with the use or maintenance of the dock equipment.
2. When a transport vehicle is parked at a loading dock, it is important that the vehicle is relatively perpendicular to the dock face and in close contact with at least one of the dock bumpers.
3. Nameplates, cautions, instructions, and posted warnings shall not be obscured from the view of operating or maintenance personnel for whom such warnings are intended.
4. Manufacturer's recommended periodic maintenance and inspection procedures in effect at date of shipment shall be followed, and written records of the performance of these procedures should be kept.
5. As with any piece of machinery, dock equipment requires routine maintenance, lubrication, and adjustments. Your local Rite-Hite representative offers owners the option of a Planned Maintenance Program (P.M.P.). As part of this service, your local Rite-Hite representative will do all routine maintenance, lubrication, and adjustments.
6. Dock equipment that is structurally damaged shall be removed from service, inspected by a manufacturer's authorized representative, and repaired as needed before being placed back in service.
7. The manufacturer shall make available replacement nameplates, caution/instruction labels, and operating/maintenance manuals upon request of the owner. The owner shall see that all nameplates, caution/instruction markings or labels are in place and legible, and that the appropriate operating/maintenance manuals are provided to users.
8. Modifications or alterations of dock equipment shall be made only with written permission of the original manufacturer. These changes shall also satisfy all safety recommendations of the original equipment manufacturer for the particular application of the dock equipment.
9. In order to be entitled to the benefits of the standard product warranty, the dock equipment must have been properly installed, maintained and operated within its rated capacities and/or specific design parameters, and not otherwise abused.
10. It is recommended that trailers equipped with air ride suspensions should remove the air from the suspension to minimize trailer bed drop, prior to loading or unloading.
11. When industrial trucks are driven on and off transport vehicles during the loading and unloading operation, the brakes on the transport vehicle shall be applied and wheel chocks or a positive restraining device shall be engaged.
12. It is recommended that an adequate stabilizing device or devices be employed at the front of the trailer in all cases where a trailer is being loaded or unloaded with the trailer resting on its support legs (landing gear) rather than a tractor fifth wheel or a converter dolly.
13. In selecting dock equipment, it is important to consider not only present requirements but also future plans or adverse environments.

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NOTICE TO USER

Definition and Function

The SHR-5000 Dok-Lok vehicle restraint is an electromechanical, self-aligning restraint device used to secure trucks and semi-trailers with an intact Rear Impact Guard (R.I.G.) to the face of a loading dock. This is achieved by securing the R.I.G. with an electrically powered steel hook. This prevents forward movement of the truck/trailer that may create an unsafe void between the face of the dock and the rear end of the truck/trailer as a forklift travels from the loading dock onto the trailer; or to create an obstruction noticeable to the truck driver, should the driver accidentally try to pull the truck/trailer away while it is being serviced.

The proper activation of the barrier and the locking mechanism is monitored by:

- Visual Control

  1 set of flashing green or red lights located at the inside of the building for dock personnel and 1 set located outside of the building for the truck driver. In addition to the lights, there are 3 instruction signs.

- Audio Control

  A horn will sound at the inside of the building, warning the forklift operator if there is not R.I.G. present, or if the engagement is improper. In this case, the trailer must be secured by other means (wheel chokes, etc.) prior to servicing trailer.

Prerequisite for proper barrier engagement is that the trailer is parked firmly against a 4in [102mm] (trade standard) thick dock bumper. The activation/deactivation is solely controlled from inside of the building by pressing the Lock (raise) button or the Unlock (lower) button.

The normal mode of the hook is in the lower STORED position, showing a flashing red light (trailer not secured) at the inside of the building and a flashing green light (trailer free to move to or away from the loading dock) at the outside of the building.

Once the trailer is parked, the dock attendant will depress the Lock button. This will raise the hook to engage the R.I.G. As soon as the R.I.G. is properly locked, there will be a simultaneous light change - the inside will change from red to green flashing (trailer secured and supported) and the outside will change from green to red flashing (**DO NOT** move trailer). After the service is complete, the dock attendant will have to depress the Unlock button which then will return the hook to its lower STORED position.

A proper hook engagement is achieved when the hook raises unobstructed to fully trap the horizontal cross member of the R.I.G. Assembly. An improper hook engagement is if the horizontal cross member of the R.I.G. is missing, obstructed or it is bent or located too far from rear of trailer that it will prevent the free passage of the hook. At this point, the trailer must be secured by other means (example: wheel chocks) in order to become serviceable.

SAFETY

Safety Identifications

DANGER

Indicates a hazardous situation which, if not avoided, will result in death or serious injury.

WARNING

Indicates a hazardous situation which, if not avoided, *could result in death or serious injury*.

CAUTION

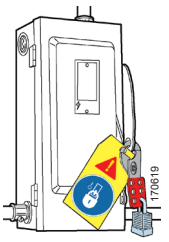
Indicates a hazardous situation which, if not avoided, *could result in minor or moderate injury*.

NOTICE

Indicates a situation which can cause damage to the equipment, personal property and/or the environment, or cause the equipment to operate improperly.

NOTE: A note is used to inform you of important installation, operation, or maintenance information.

Lockout Procedure



Barricade work area and post safety warnings.

Power supply/control must:

- Be disconnected or locked in OFF position using a lockout device approved by local codes.
- Have signage that:
  - Clearly states repairs are being made.
  - Identifies person responsible for lockout condition.

    NOTE: Only this person should be able to remove warnings and lockout device.
  - Withstands environmental conditions (weather, wet, and damp, etc.) and remains readable.

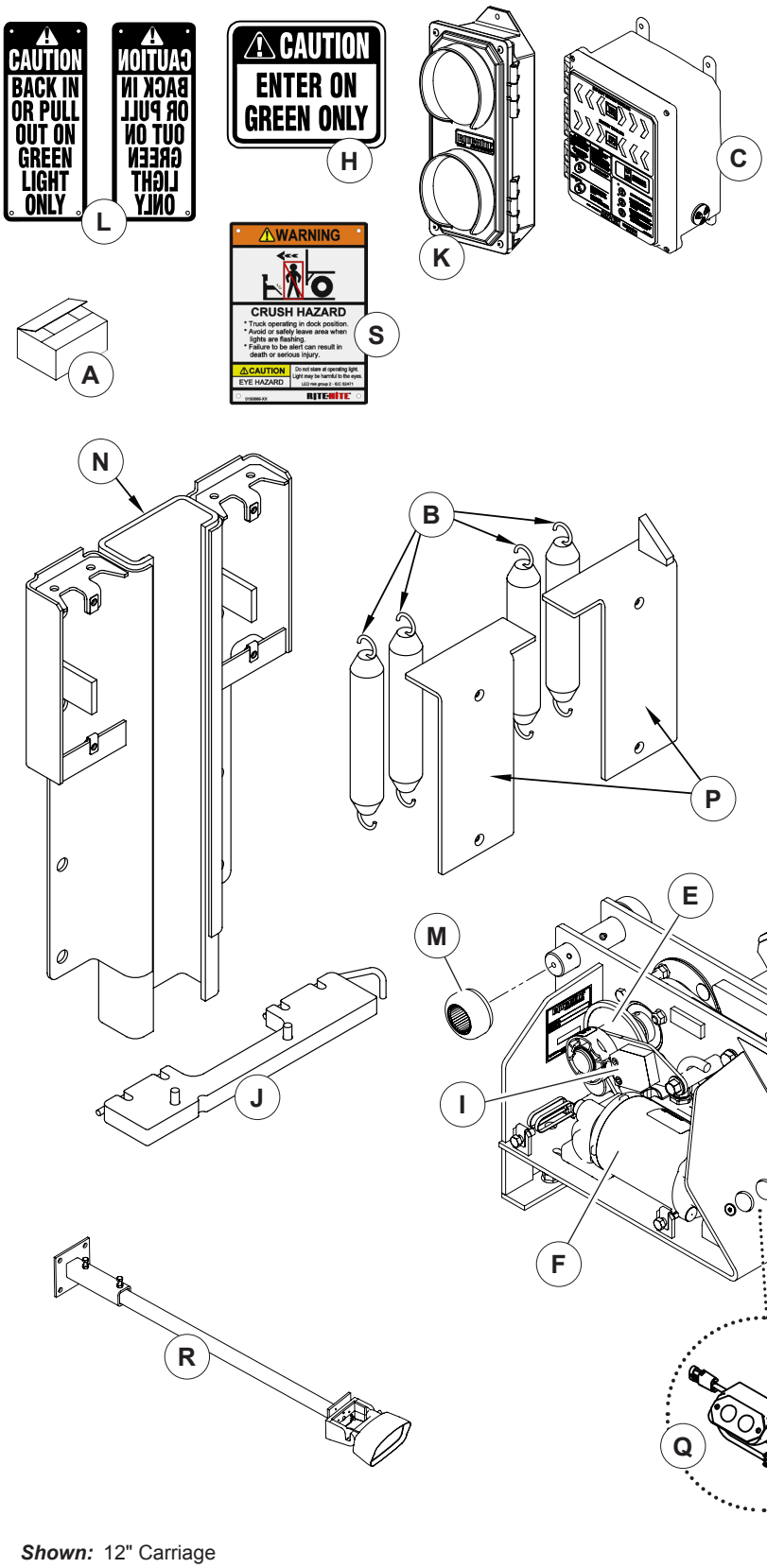
General

DANGER

A qualified electrician should install wiring in accordance with local electrical codes.

Use lockout procedures to prevent death or severe personal injury.

FEATURES



A. Anchor Kit
B. Carriage Return Springs
C. Control Box
D. Control Harness
E. Drive Gear
F. Electric Motor Assembly
G. Hook
H. Inside Sign
I. Limit Switches
J. Lower Spring Bar
K. Outside Lights
L. Outside Signs
M. Roller Assembly
N. Roller Track Assembly
O. Slope Extension
P. Spring Covers

Approach-Vu™ (Optional):

Q. Light and Alarm
R. Truck Sensor
S. Approach-Vu Sign

Shown: 12" Carriage

Figure 1

ELECTRICAL

Control Harness

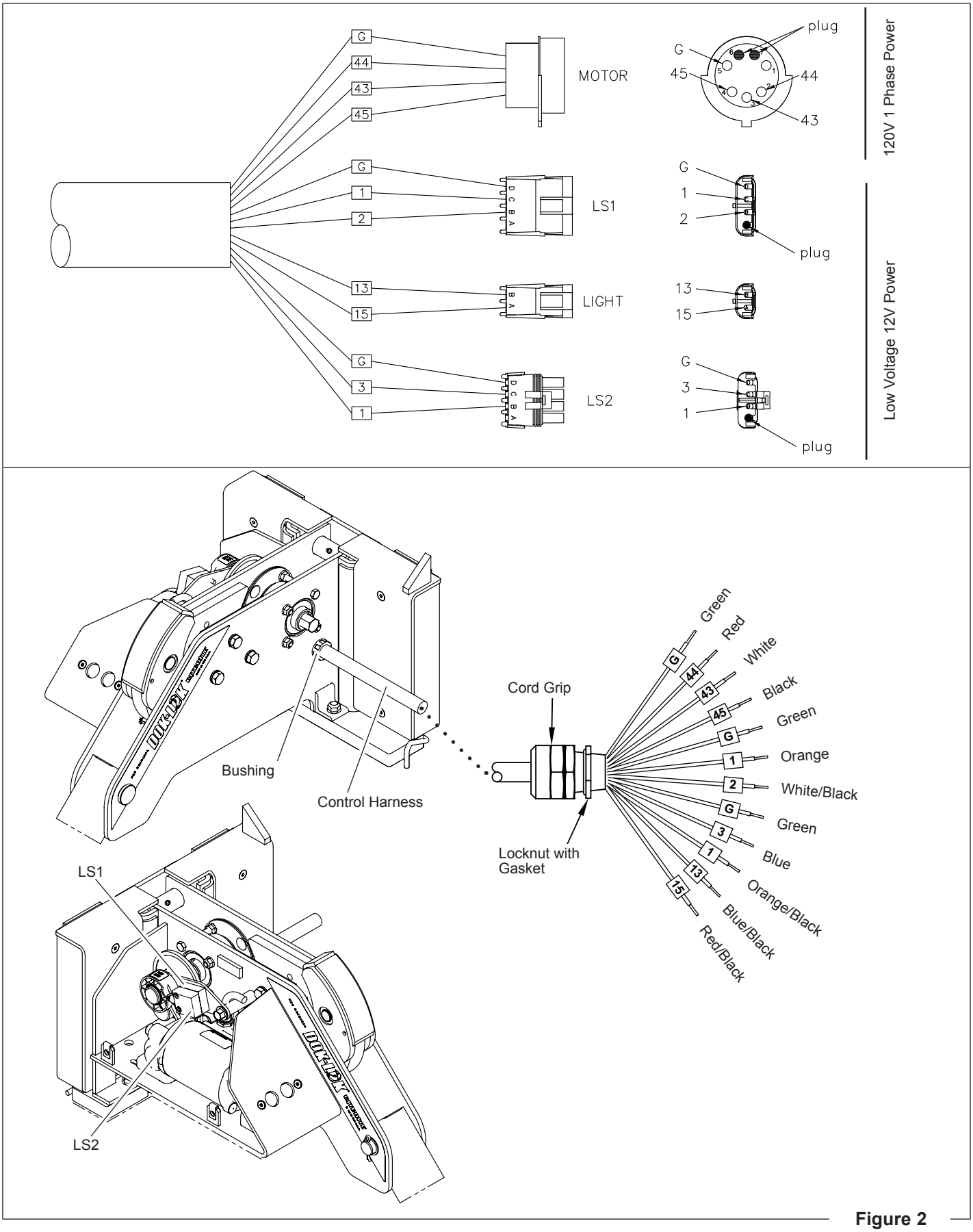


Figure 2

ELECTRICAL

Power-Board

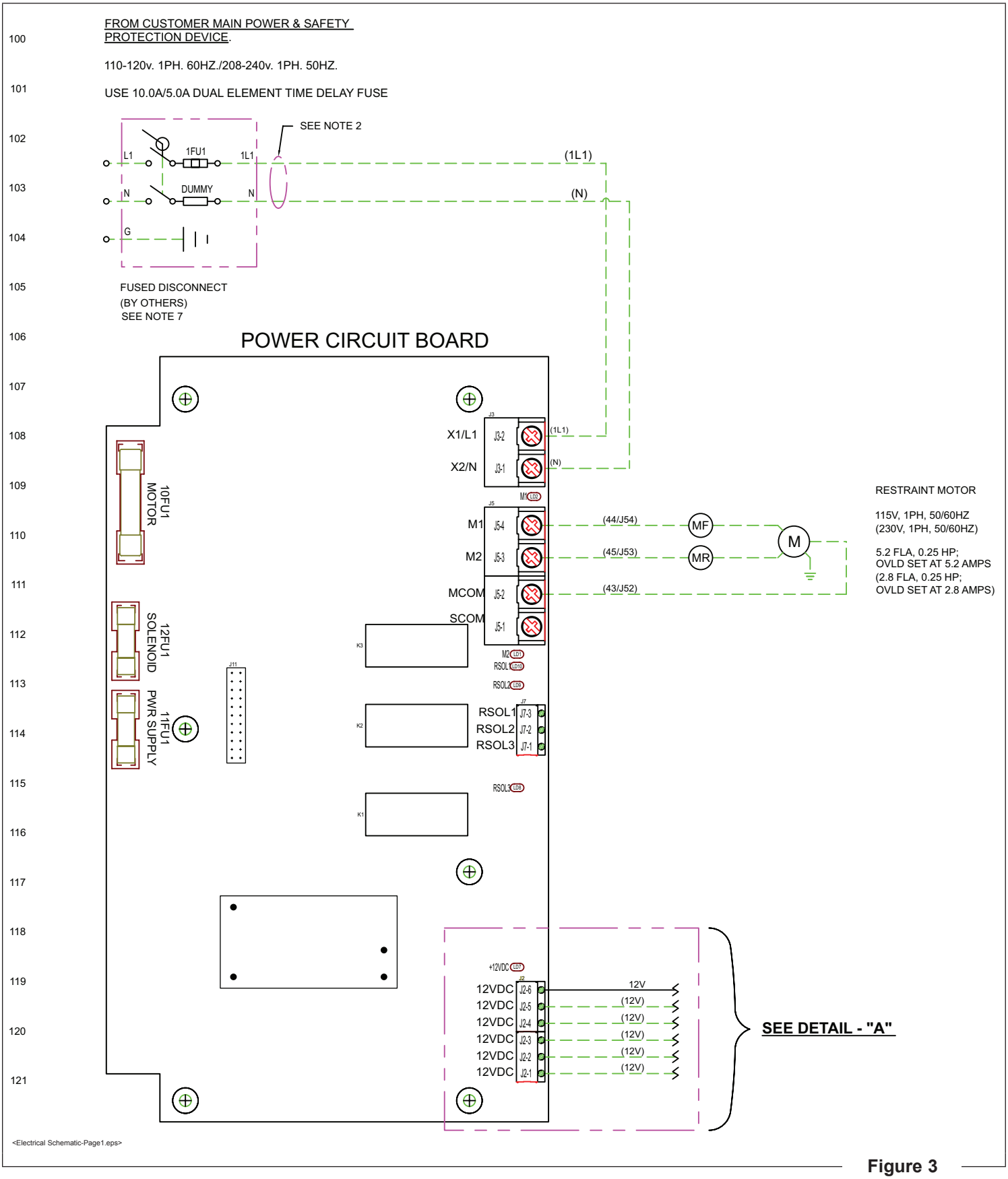


Figure 3

ELECTRICAL

Restraint Micro Board

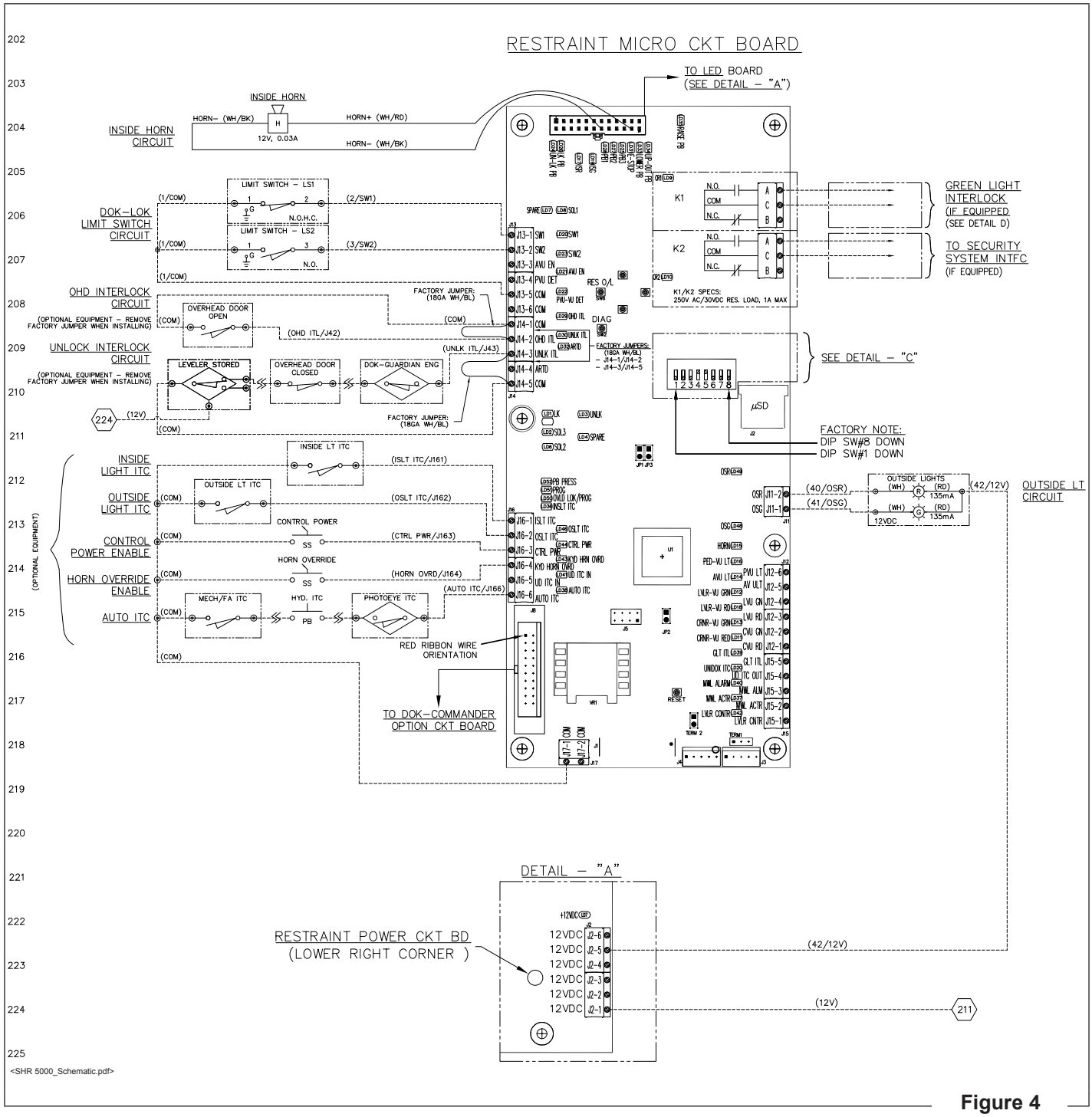
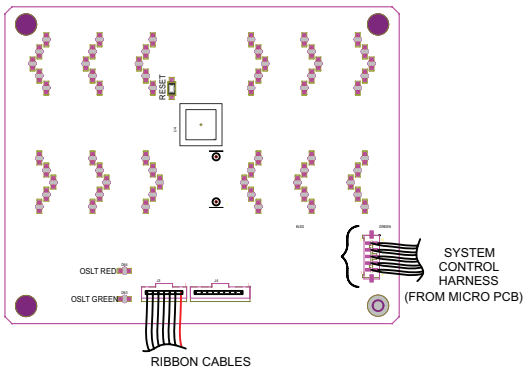


Figure 4

ELECTRICAL

Restraint Micro Board *Continued*

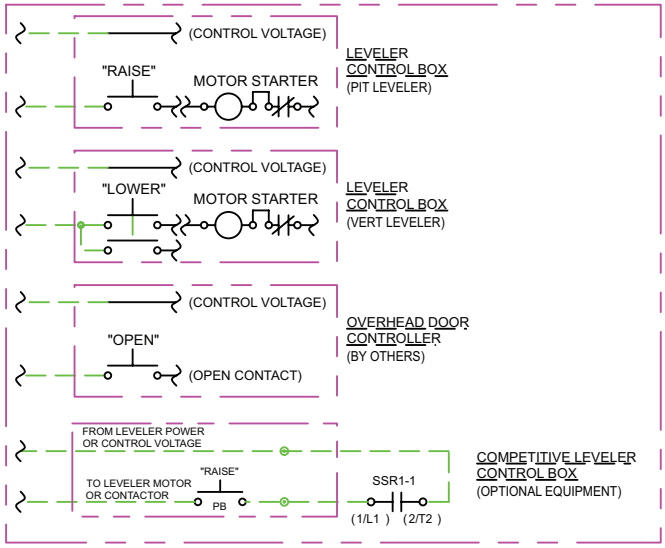
DETAIL B - BACK OF COVER



System Control Harness - MEMBRANE

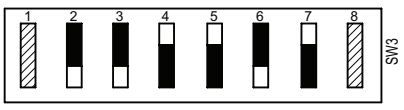
uControl Board		LED Board		
J7	Description	J1	Description	Wire Color
1	USART2_RX	1	MICROPCB_RX	BLUE
2	USART2_TX	2	MICROPCB_TX	GRAY
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13	12V [Vext]	5	12V [Vext]	RED
14	OSG INDICATOR	9	OSG INDICATOR	WHITE/GREEN
15	OSR INDICATOR	10	OSR INDICATOR	WHITE/ORANGE
16	ISR	8	ISR	ORANGE
17	ISG	7	ISG	GREEN
18				
19	HORN [-]			WHITE/BLACK
20	GND	4	DIGITAL GND	WHITE/BLUE
21	RESERVED	6	RESERVED	BLACK
22	USART2_RTS	3	MICROPCB_RTS	WHITE
23	12V [HORN +]			
24				

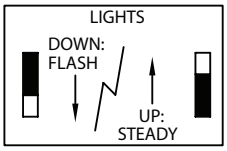
DETAIL - "D" GREEN LIGHT INTERLOCK VARIATIONS



DETAIL - "C" MICRO CONTROL BOARD DIP SWITCH SETTINGS

DIP SWITCH	FACTORY SETTING	OPTION/FUNCTION
1	DOWN	FACTORY USE
2	DOWN	INSIDE RED LIGHT
3	DOWN	INSIDE GREEN LIGHT
4	UP	CORNER & LEVELER-VU RED
5	UP	CORNER & LEVELER-VU GREEN
6	DOWN	OUTSIDE LIGHTS - RED/GREEN
7	UP	PEDESTRIAN-VU
8	UP	CODED HORN OVERRIDE
	DOWN	STANDARD HORN OVERRIDE





<Electrical Schematic-Page3.eps>

1. INSTALL PER LOCAL ELECTRICAL CODES. REFER ALL INSTALLATION AND SERVICE TO QUALIFIED PERSONNEL.
2. ALL INCOMING POWER (FROM DISCONNECT TO CONTROL BOX) AND MOTOR FIELD WIRING TO BE MINIMUM #14GA. 60°/75°C COPPER WIRE, INSULATED SUFFICIENTLY FOR INCOMING VOLTAGE.
3. ALL CONTROL FIELD WIRING TO BE MINIMUM #14GA. 60°/75°C COPPER WIRE ONLY, INSULATED SUFFICIENTLY FOR INCOMING VOLTAGE.
4. ALL INTERNAL WIRING TO BE #14GA. MINIMUM FOR POWER CIRCUITS AND CIRCUIT BOARD POWER, #16GA MINIMUM FOR CONTROL CIRCUITS, 90°C, RED COPPER WIRE UNLESS OTHERWISE NOTED, INSULATED SUFFICIENTLY FOR INCOMING VOLTAGE.
5. TORQUE REQUIREMENTS:
   -PCB TERMINAL BLOCK POWER (J3/J5): 7.0 LB-IN
   -PCB TERMINAL BLOCK ALL OTHER: 4.0 LB-IN
   -GROUND TERMINAL: 35 LB-IN
6. LS1 & LS2 PROVIDED BY RITE-HITE WITH VEHICLE RESTRAINT. OTHER FIELD DEVICES ARE OPTIONAL. SEE SPECIFIC ELECTRICAL SCHEMATIC.
7. FUSED DISCONNECT NOTES:
   -FUSED DISCONNECT IS NOT PROVIDED BY RITE-HITE PRODUCTS CORPORATION. DISCONNECT MUST BE PROVIDED BY OTHERS AND INSTALLED PER LATEST EDITION OF UL508A AND NEC REQUIREMENTS.
   -A BRANCH CIRCUIT DISCONNECT SHALL BE LOCATED WITHIN A 50 FT. RADUIS AND BE VISIBLE FROM THE CONTROL BOX LOCATION. [REFERENCE LATEST EDITION OF NEC, SECTION 430]
8. CLASS 1 CONTROL CIRCUIT.

LEGEND:
- DENOTES WIRE CONNECTIONS THRU TERMINAL BLOCK.
- DENOTES FIELD WIRES.
- DENOTES WIRE NUT CONNECTION.
- DENOTES MALE/FEMALE PLUG CONNECTOR.

Figure 5

ELECTRICAL

Wiring: Corner-Vu / Leveler-Vu

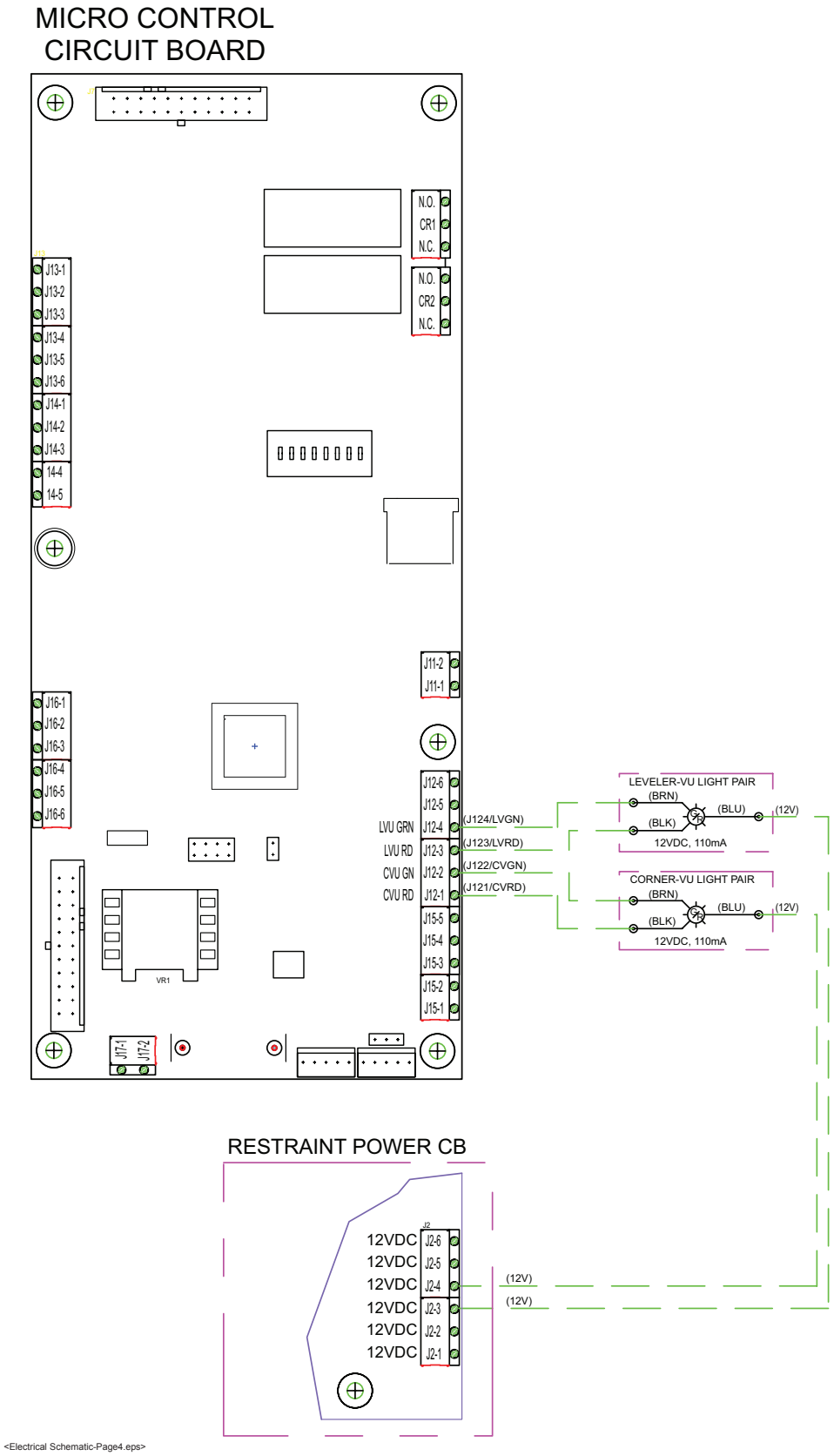
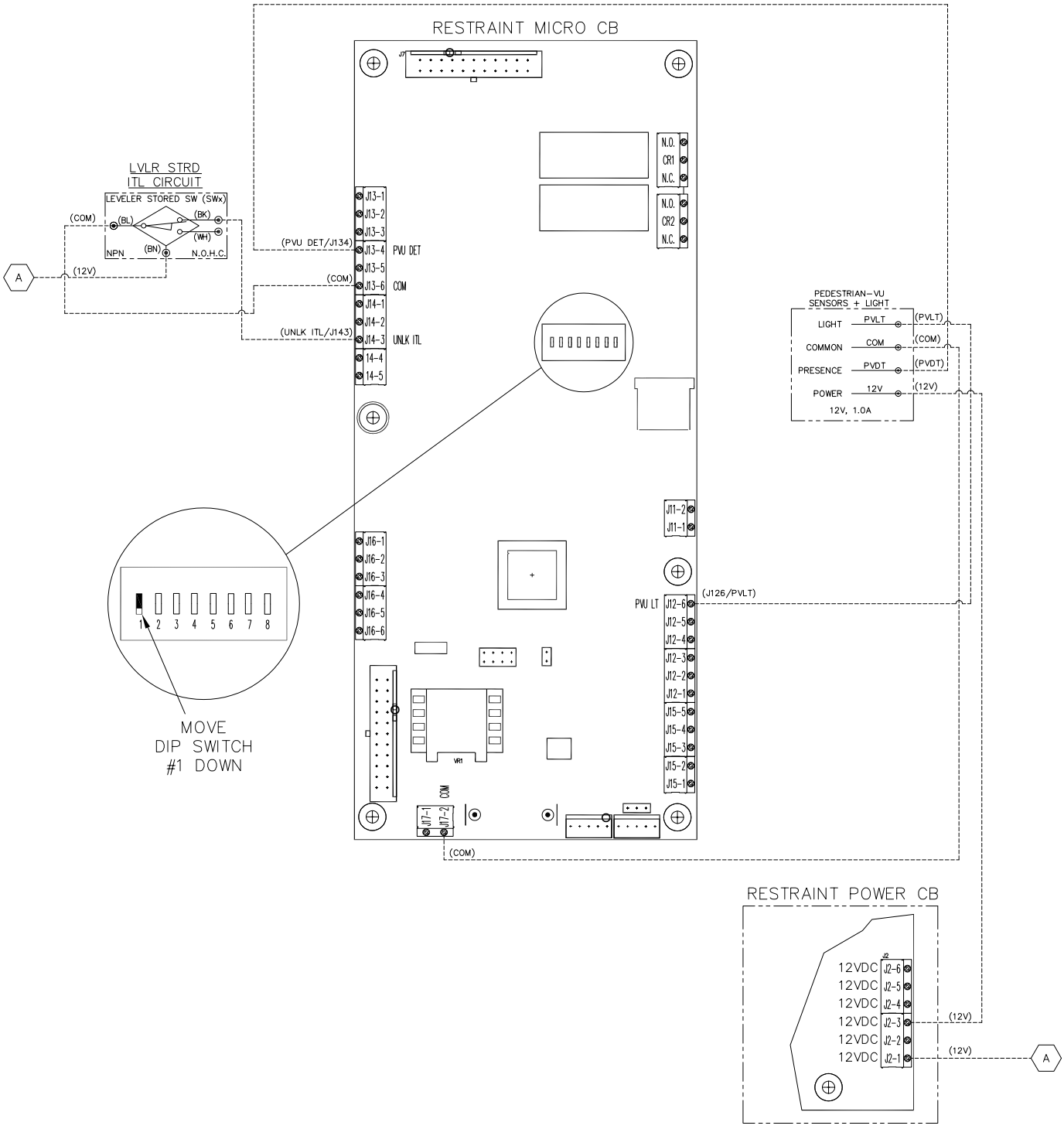


Figure 6

ELECTRICAL

Wiring: Pedestrian-Vu

Leveler Stored to Unlock Restraint



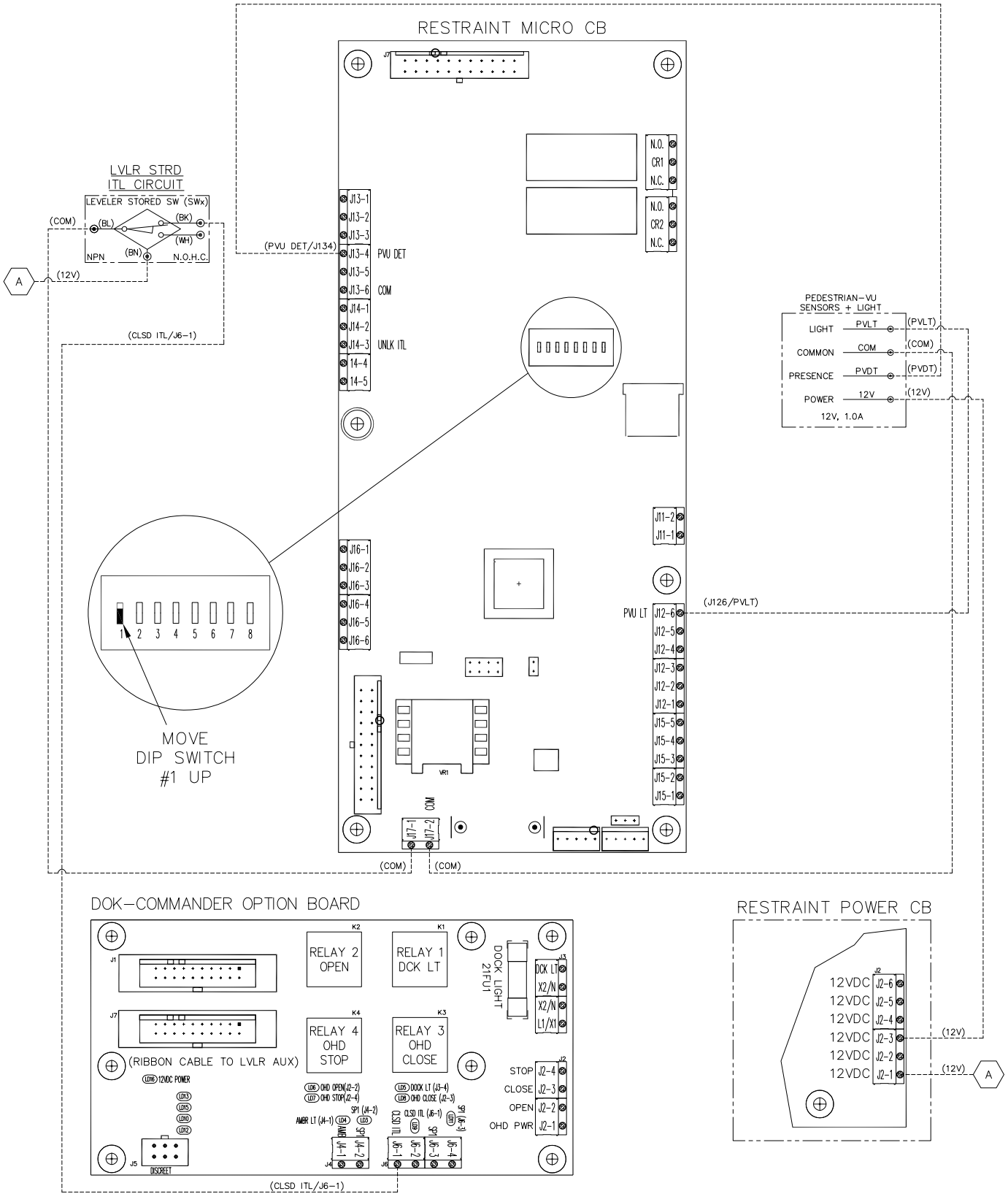
<Rite-Vu_PED-VU STRD UNLK Schematic.PDF>

Figure 7

ELECTRICAL

Wiring: Pedestrian-Vu *Continued*

Leveler Stored to Close Overhead Door

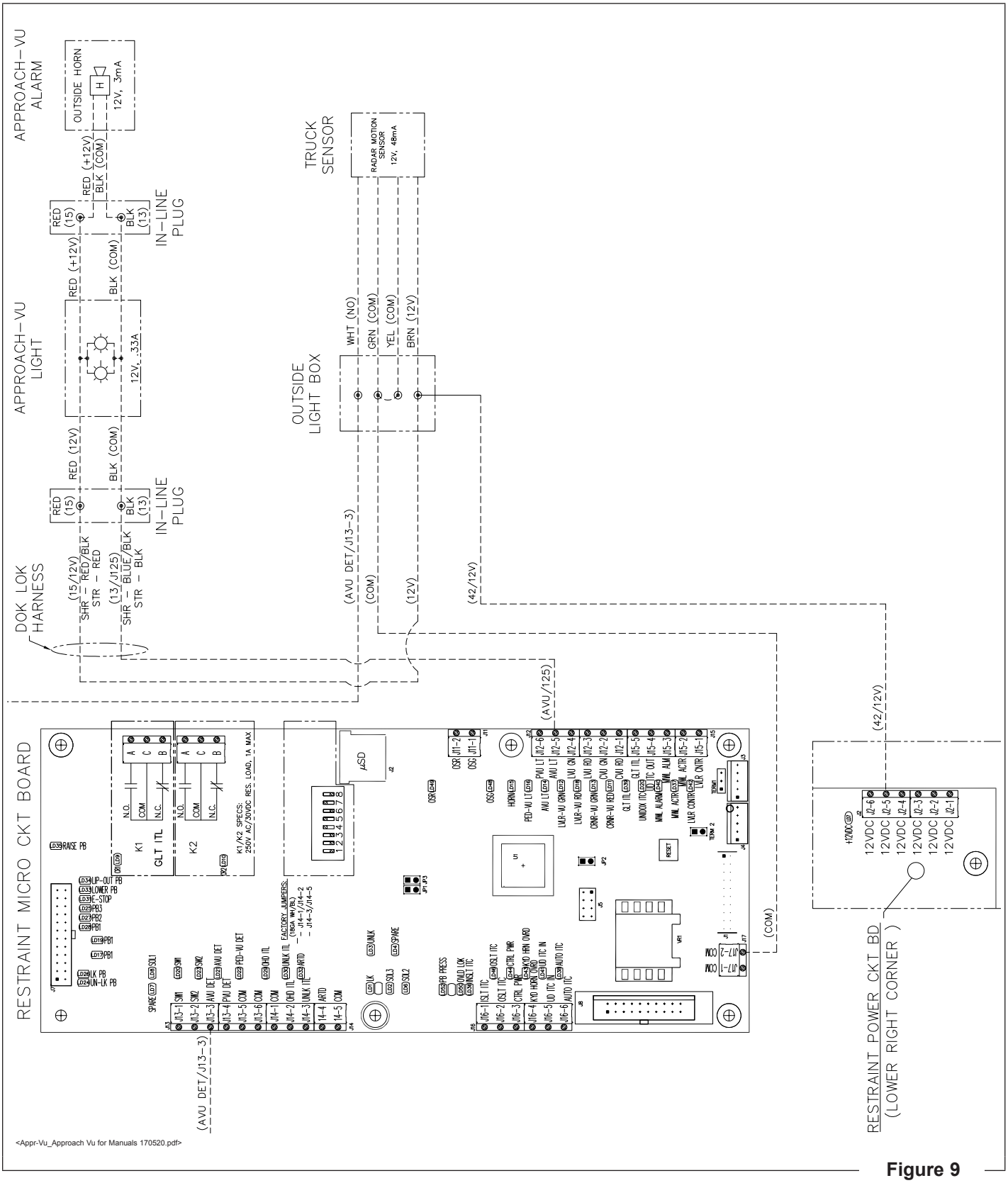


<Rite-Vu_PED-VU OHD STRD UNLK Schematic.PDF>

Figure 8

ELECTRICAL

Wiring: Approach-Vu™



<Appr-Vu_Approach Vu for Manuals 170520.pdf>

Figure 9

ELECTRICAL

Wiring: Outside Light Box

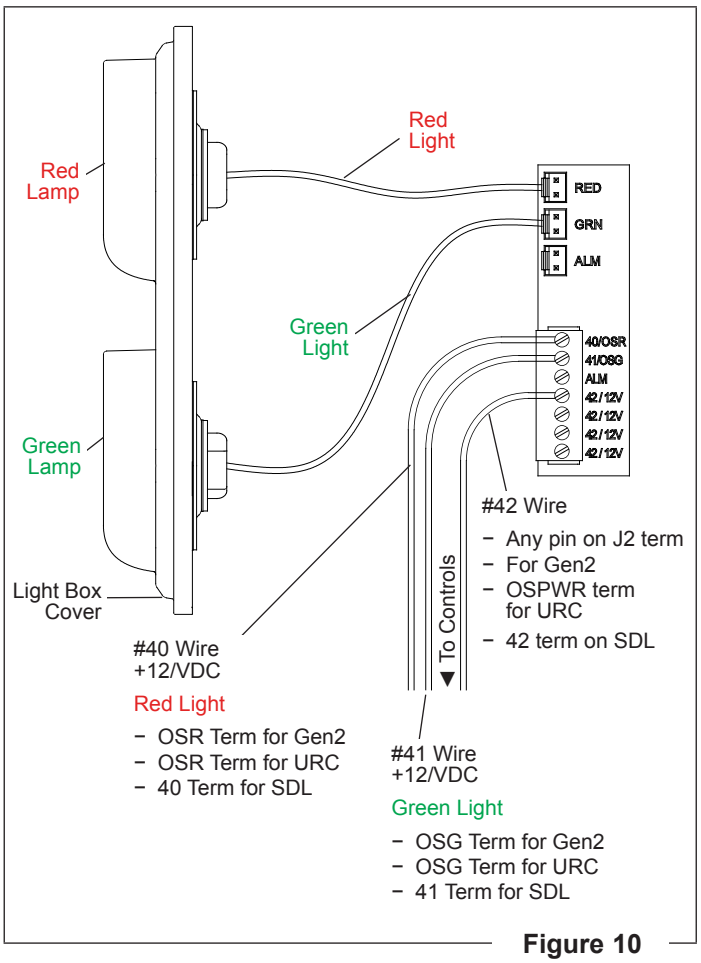


Figure 10

OPERATION

⚠ WARNING

Before loading or unloading a vehicle at your loading dock while using a Dok-Lok vehicle restraint, always visually confirm barrier blocks R.I.G. assembly. Secure trailer by other means, if a condition occurs that cannot be fixed by backing trailer firmly against dock bumpers.

Area around R.I.G. assembly must be free of plates or other obstructions.

Always operate Dok-Lok vehicle restraint from top of dock.

Dok-Lok vehicle restraints should be operated by authorized personnel who have read and understand Owner's Manual.

Contact your local Rite-Hite representative or Rite-Hite (U.S.) 800-456-0600 with any questions.

Failure to follow these procedures could allow unexpected trailer / loading dock separation, resulting in death or serious injury.

⚠ WARNING

Power moving components. Serious injury could occur.

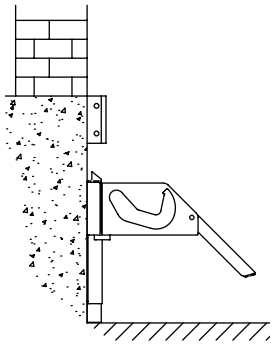
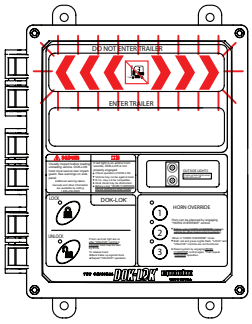
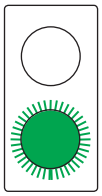
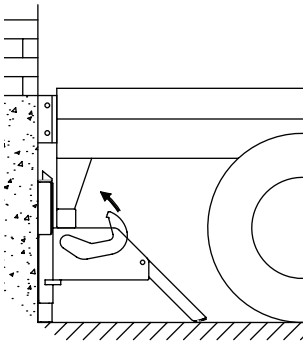
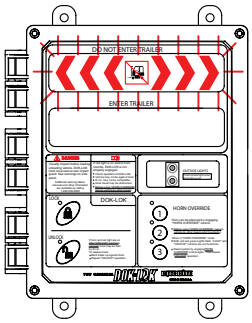
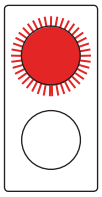


Slip hazard: Do not use equipment as a step.

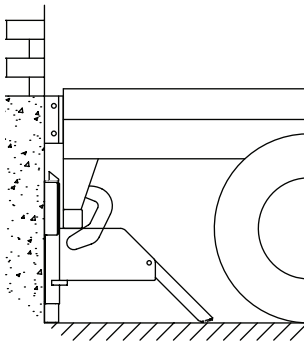
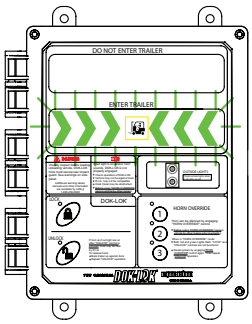
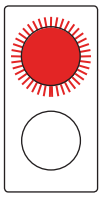
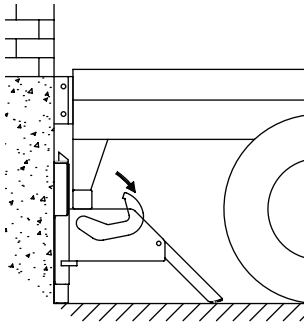
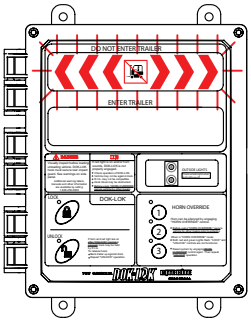
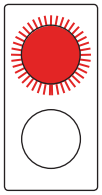


Pinch hazard: Keep all extremities away.

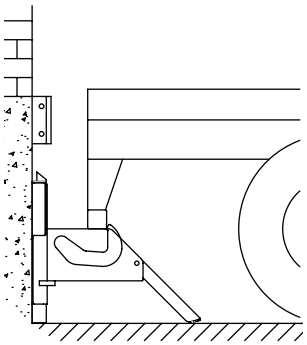
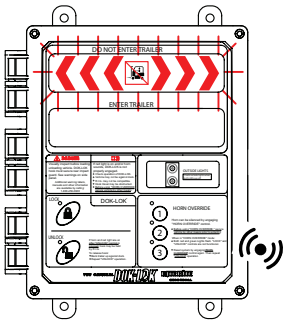
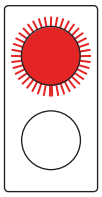
OPERATION

RESTRAINT		DEFAULT STATUS	
STATE	**POSITION**	**CONTROL BOX: INSIDE LIGHTS, SOUND**	**OUTSIDE LIGHTS**
Restraint UNLOCKED/ STORED position. Hook is in STORED position.		 Red (flashing) Loading or unloading is not permitted, since R.I.G. is not locked.	 Green (flashing) A vehicle may pull into or out of loading bay.
Restraint LOCKING (LOCK button pressed) • Trailer has backed into loading dock and is parked firmly against dock bumpers. • Hook rotates from STORED position to entrap R.I.G. If horn sounds, go to FAULT State from LOCKING State if not, go to Restraint LOCKED. **⚠ CAUTION** **If trailer can not be restrained due to a lift gate or other obstruction that could become damaged, go to HORN OVERRIDE State.**		 Red (steady) Alerts operator that an unsafe condition exists while hook and support cylinders are in transit.	 Red (flashing) Alerts truck driver not to move.

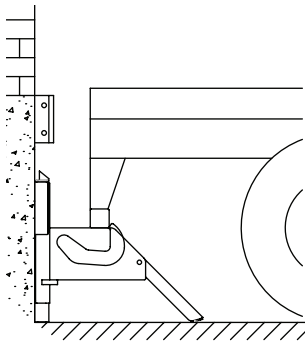
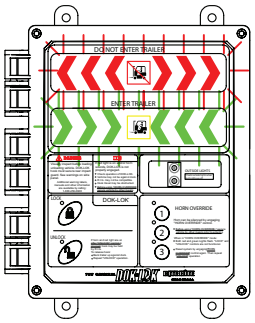
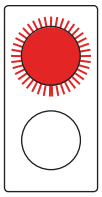
OPERATION

RESTRAINT		DEFAULT STATUS	
STATE	POSITION	CONTROL BOX: INSIDE LIGHTS, SOUND	OUTSIDE LIGHTS
Restraint LOCKED When R.I.G. is entrapped by hook, a LOCKED condition exists. If during loading/unloading inside light turns red and horn sounds, press LOCK button to secure R.I.G. **⚠ WARNING** **Visually confirm Dok-Lok barrier obstructs R.I.G. of trailer being serviced, before operating dock leveler.**		 Green (flashing) Alerts operator a safe condition exists.	 Red (flashing) Alerts truck driver not to move.
Restraint UNLOCKING (UNLOCK button pressed) Hook travels from LOCKED position to STORED position. If horn sounds go to FAULT State from UNLOCKING State.		 Red (steady) Alerts operator that an unsafe condition exists while hook and support cylinders are in transit.	 Red (flashing) Alerts truck driver not to move.

OPERATION

RESTRAINT		DEFAULT STATUS	
STATE	**POSITION**	**CONTROL BOX / INSIDE LIGHTS**	**OUTSIDE LIGHTS**
FAULT State from LOCKING State Hook cannot entrap R.I.G. – R.I.G. could be located too far from rear of trailer, bent, obstructed or missing. If trailer is parked firmly against dock bumpers go to HORN OVERRIDE state. If not, press UNLOCK to clear fault, have trailer back up and repeat Restraint LOCKING.		 Red (flashing) Horn (pulsing) Alerts operator that trailer is not locked.	 Red (flashing) Alerts truck driver not to move.
FAULT State from UNLOCKING State Hook cannot rotate to STORED position. – Hook could be caught on R.I.G. or another part of trailer. Verify trailer is parked firmly against dock bumpers. If not, press LOCK to entrap R.I.G., have trailer back up and repeat Restraint UNLOCKING.			

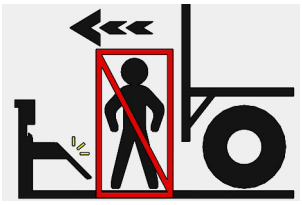
OPERATION

RESTRAINT		DEFAULT STATUS	
STATE	**POSITION**	**CONTROL BOX / INSIDE LIGHTS**	**OUTSIDE LIGHTS**
HORN OVERRIDE State **(HORN OVERRIDE code entered after securing trailer by alternate means)** An alternate means of securing truck must be used if hook can not capture R.I.G. (i.e. wheel chocks). To return to STORED: 1. Enter HORN OVERRIDE code ***NOTE:*** Factory preset = 1223 2. Press UNLOCK button. **DANGER** **Before operating HORN OVERRIDE, secure trailer by other means.**		 Red (flashing) Green (flashing) Alerts operator trailer is secured by other means.	 Red (flashing) Alerts truck driver not to move.

OPERATION

Approach-Vu™ (OPTIONAL)

⚠ WARNING



CRUSH HAZARD

- Truck operating in dock position.
- Avoid or safely leave area when lights are flashing.
- Failure to be alert can result in death or serious injury.

⚠ CAUTION

Do not stare at operating light. Light may be harmful to the eyes.

Truck Detection Zone

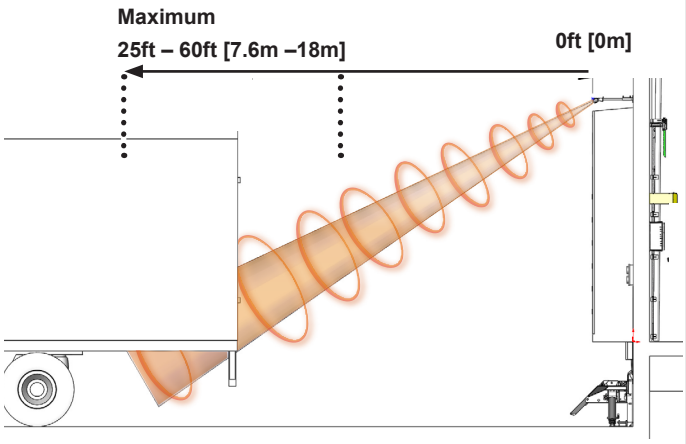


Figure 11

Hazard Notification on Restraint

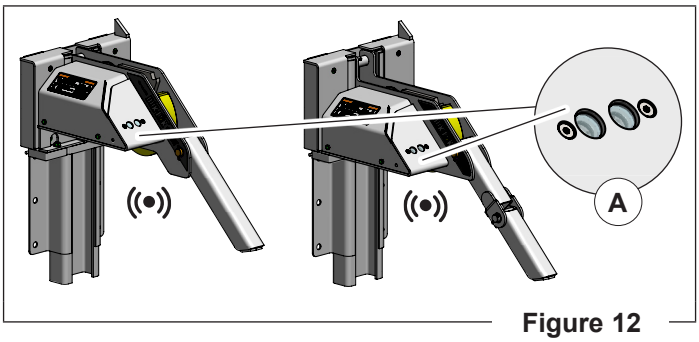
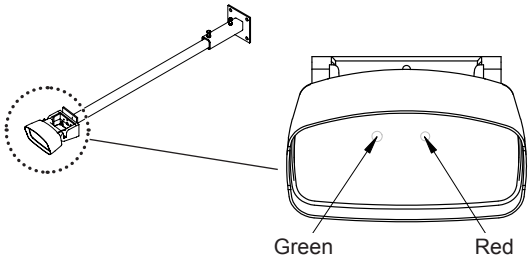


Figure 12

When a truck is detected:

- Lights will flash (A).
- Alarm located in restraint will sound.

Truck Sensor LEDs



STATE	LEDS
Start-Up	Red/Green (Flashing)
Ready	Green (Steady)
Detects Truck	Red (Flashing)
False Detection	Green or Red/Green (Flashing)

Figure 13

MAINTENANCE

DANGER

When working with electrical or electronic controls, make sure that power source has been locked out and tagged according to approved local electrical codes.

Post safety warnings and barricade work area, at dock level and approach level, to prevent unauthorized use of dock position.

WARNING

A safe work place requires all lights and horn to be working properly. DO NOT use Dok-Lok vehicle restraint if parts are broken or missing.

CAUTION

When lifting unit use a proper lifting device (crane, jack). Lifting by hand may cause back injury.

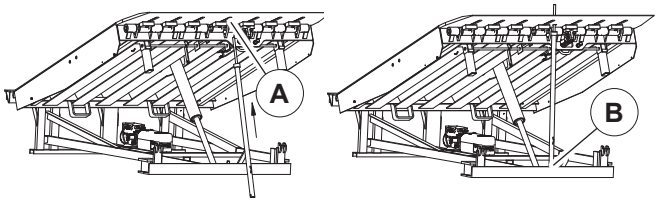
NOTICE

Contact Rite-Hite if loading docks are exposed to harsh environments (extreme climates, corrosive chemicals, high usage, etc.). More frequent maintenance may be required.

A Planned Maintenance Program (P.M.P.), customized to your specific operation is available and recommended. For a P.M.P., contact your local Rite-Hite representative.

NOTE: If a leveler is installed at the Dok-Lok vehicle restraint location, it may be necessary to raise the leveler before performing maintenance.

DANGER



When working under dock leveler platform or lip:

1. **Install Safe-T-Strut™ with the assistance of another person.**
   - Raise leveler until platform reaches its highest position and lip extends. Maintain this position.
   - (A) Have assistant insert the smaller end of the Safe-T-Strut™ through hole in the middle of leveler lip and place strut's wider open end over the base located on leveler's front frame (B).
   - Align the holes on the base and Safe-T-Strut™ so leveler may be secured with retaining pin and safety clip.
   - Release the pushbutton on powered levelers allowing the Safe-T-Strut™ to rest on the underside of lip.

   NOTE: If you are unable to install the maintenance support, contact your Rite-Hite Service Representative or Rite-Hite Customer Service at 1-414-355-2600.
2. **Lockout and tag power supply.**
   - Turn off power to control box.
   - Lock out and tag the main power source
   - Barricade leveler at dock and approach levels to prevent unauthorized use.

Safe-T-Strut™ removal:

- Have an assistant raise leveler to its highest position with lip fully extended.
- Release safety clip and remove retaining pin.
- Lift the support off base.
- Return Safe-T-Strut™ to it's storage position.

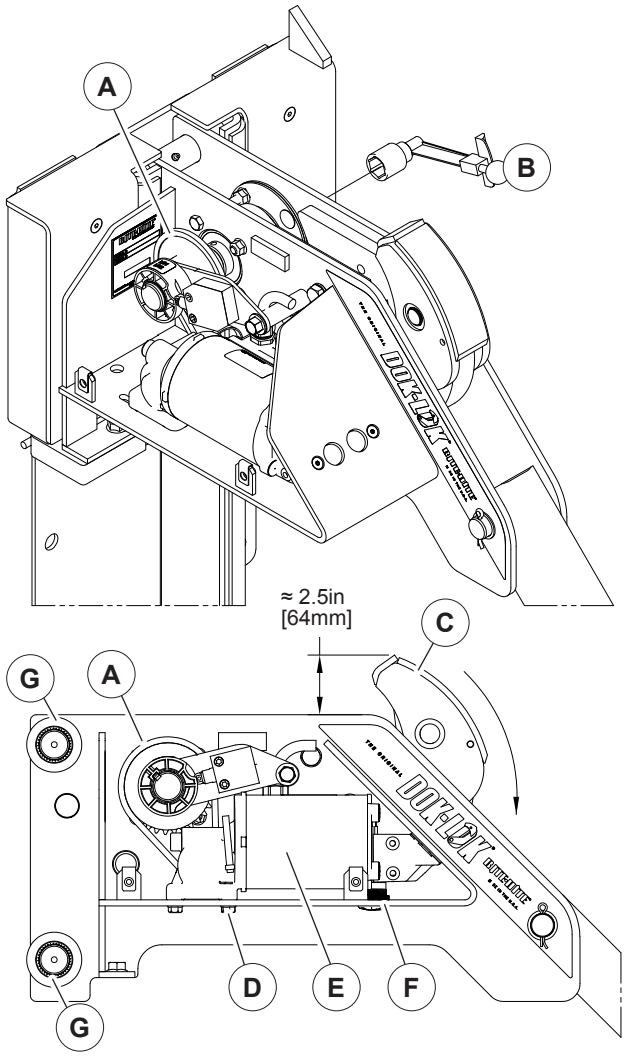
MAINTENANCE

Planned Maintenance

#	P.M. IN DAYS 1	180	360	INSPECT AND PERFORM:
1	✓	✓	✓	Remove debris around Dok-Lok vehicle restraint
2	✓	✓	✓	Verify horns, inside and outside lights are working.
3	✓	✓	✓	Replace damaged or missing light bulbs and lenses.
4	✓	✓	✓	Repair, remount, or replace outside and inside signs as required.
5	✓	✓	✓	Inspect dock bumpers. 4in [102mm] of protection is required. Worn, torn, loose or missing bumpers must be replaced.
6	–	✓	✓	See **"Figure 14".** Grease rollers (G) at fittings (top and bottom axle). Grease hook at fittings. Use Mobilith SHC220 No. 2 grease (or equivalent temperature range lithium based grease). 1st 180 days: 7–8 pumps 180 Day intervals (after the first 180): 2–3 pumps ***Note:*** Items that use SAE 30 weight oil lubricant only require lubrication every 360 Days.
7	–	✓	✓	Check that all concrete anchor bolts are torqued to 60 ft-lbs.
8	–	✓	✓	Verify clutch torque at the hook shaft is greater than 500 in-lbf.
9	–	✓	✓	Inspect outside junction and light box. They should be securely mounted. If loose or damaged, inspect all wires and wire connections.
10	–	✓	✓	Inspect flexible electrical harness from Dok-Lok vehicle restraint to junction box. Look for kinks, crushed areas, etc.
11	–	✓	✓	Verify hook is centered between stop plates.
12	–	✓	✓	Inspect all conduit boxes, control boxes and electrical connections for damage. Repair or replace if worn or damaged. If control box has evidence of condensation: a. Inspect conduit. Conduit should be routed to enter through the bottom or side of the enclosure. A drip leg may be needed if the conduit is filling with water. b. Inspect the seal on the cover of the enclosure. The seal should be securely adhered to the cover with no signs of peeling or bubbling. Repair or replace if worn or damaged. c. For non-metallic enclosures, breather vent (part number 122130) may be installed. The vent is NEMA 4X and will not change the environmental rating of the control box.
13	–	✓	✓	Perform operational test after all maintenance repairs and adjustments are complete.
14	–	–	✓	Check and tighten control box mounting hardware
15	–	–	✓	Drive Chain: Check lubrication and tightness. See **"Drive Chain" on page 24**
	Lubrication may be required more frequently, based on usage and environmental conditions.			

MAINTENANCE

Drive Chain

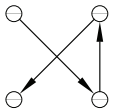


A. Drive Chain
B. Torque Wrench Flex-beam style (supplied by others)
C. Hook position to check torque
D. Mounting Bolts
E. Motor
F. Motor Mounting Plate
G. Fittings

Figure 14

Inspect the drive chain (A) for dirt and chain tightness:

1. Open the motor enclosure.
2. Clean the chain with solvent.
3. Spray the chain with chain spray type lubricant (Zep 2000 or equivalent).
4. Verify clutch torque at the hook shaft:
   a. Start with hook in position (C) with tip of hook ≈ 2.5in [64mm] above carriage side plates.
   b. Use a flex-beam style torque wrench (B) to rotate hook shaft/hook down toward stored position, to test the torque. Record the maximum torque achieved just before clutch slips.
   c. If the clutch torque measured is less than 500 in-lbs, gear motor should be replaced.
5. The drive chain should have 1/8in [3mm] to 1/4in [6mm] slack. To tighten chain:
   a. Open motor enclosure and loosen the 4 motor mounting bolts (D).
   b. Pry the motor assembly forward, in the slotted holes, until the chain is tight (with proper alignment between sprockets).
   c. Hold in this position and tighten each bolt to 105 in-lbs in increments of 10 in-lbs. Tighten in this pattern:



   d. Test for proper chain alignment and tension.
6. Close motor enclosure.

TROUBLESHOOTING

Component Testing

Limit Switch

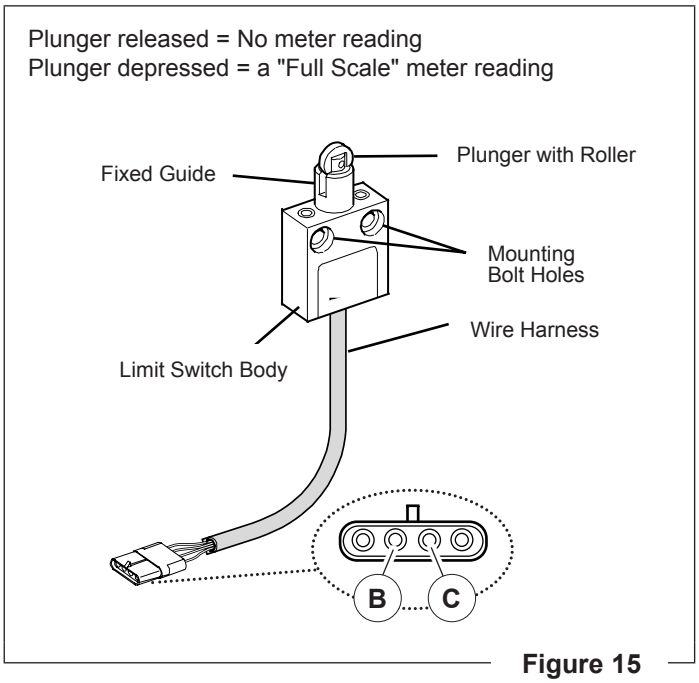


Figure 15

1. Set multimeter to “RX1” scale for “Continuity Test”.
2. Attach multimeter leads to pins (B) and (C) wires of switch connector:

Green (ground) wire of limit switch does not have to be tested. A continuity test lamp may be used instead of a multimeter.

Limit Switch and Hook Positions

POSITION					
HOOK	LS1	LS2	INSIDE LIGHT	OUTSIDE LIGHT	CONTROL BOX HORN
1. Stored	Closed / Depressed	Open / Released	Red	Green	Off
2. No Hook (Lower)	Closed / Depressed	Closed / Depressed	Red	Red	On
3. Hooked (90°)	Open / Released	Closed / Depressed	Green	Red	Off
4. No Hook (Upper)	Closed / Depressed	Closed / Depressed	Red	Red	On

Vehicle Restraint Motor

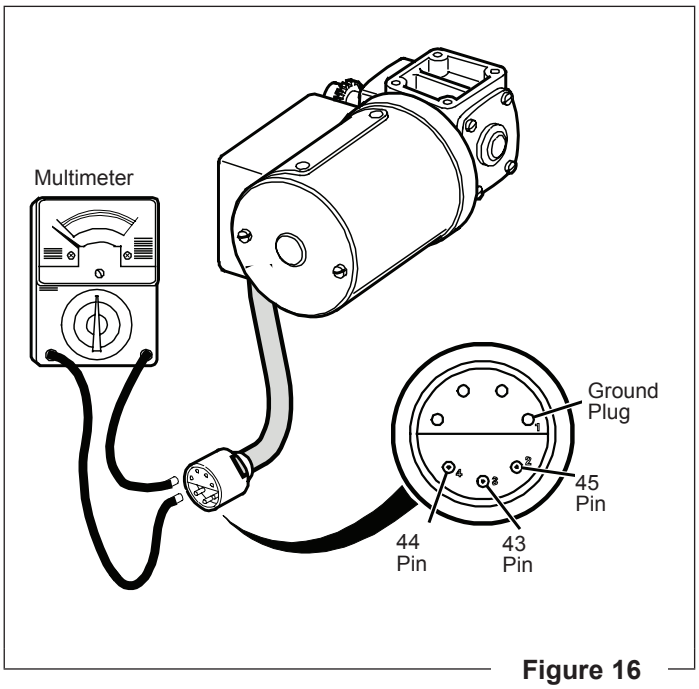


Figure 16

BadO/L:

Little or infinite ohm reading (no needle movement) between pins 43 and 44, 43 and 45. Set multimeter to ohms.

Standard readings:
Between pins 44 and 45 = 11 ohms
Between pins 44 or 45 and 43 = 5.5 ohms.

Open Winding:

Infinite ohms (no needle movement) between pins 44 and 45. Check between pins 44 and 43 or 45 and 43 to determine which winding is open.

Mechanical Binding:

Motor hums. Motor leads show continuity between all windings. Shaft does not move. Reads 6.5 amps at pin 43 on both rotations.

TROUBLESHOOTING

LED Status Chart

? = Varies (per operation)
* = On for 1.5 sec before M2 energizes
A = Alternating
D = Double fast pulsing / flashing
F = Off
ITL = Interlock input on
K = Continuous slow chirp
M = Lights when button pressed
P = Pulsing / Flashing (set to steady using DIP switches)
T = Steady On

				MICRO CONTROL BOARD																			POWER BOARD		
				INPUTS						OUTPUTS															
				FIELD			PUSH BUTTONS			12VDC									RELAY			115/230VAC		
				Dok-Lok Limit Switch 1 [SW1]	Dok-Lok Limit Switch 2 [SW2]	Unlock Interlock [UNLK ITL]	Lock	Unlock	Horn Silence (1/2/3)	Inside Red Light [ISR]	Inside Green Light [ISG]	Corner-Vu Red Light [CVU RD]	Corner-Vu Green Light [CVU GRN]	Leveler-Vu Red Light [LVU RD]	Leveler-Vu Green Light [L-VU GRN]	Outside Red Light [OSR]	Outside Green Light [OSG]	Dok-Lok Horn [HORN]	Restraint Overload LED [YELLOW]	K1 - Green Light Interlock	K2 - Security System Interface [IF EQUIPPED]	Motor Ouptut #1 [M1/LOCK]	Motor Ouptut #2 (M2/UNLOCK)	12VDC Power Supply OK
TERMINAL BLOCK NO.				J13.1	J13.2	J14.3	Membrane			J7.16	J7.17	J12.1	J12.2	J12.3	J12.4	J11.2	J11.1	J7.19	N/A	J9.3	J10.3	J5.4	J5.3	J2.1-6
POWER BOARD LEDS				–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	LD2	LD1	LD7
MICRO CONTROL BOARD LEDS				LD20	LD23	LD30	LD52			LD17	LD19	LD11	LD13	LD18	LD12	LD49	LD48	LD15	LD50	LD9	LD10	LD1	LD3	–
NO.	01.01.00	State/Sequence No.	Locked State	F	T	?	–	–	–	F	P	F	P	F	P	P	F	F	F	T	T	F	F	T
	01.01.01		Locking Sequence	T	T	?	M	–	–	T	F	P	F	P	F	P	F	F	F	F	T	T	F	T
	01.02.00		Unlocked State	T	F	?	–	–	–	P	F	P	F	P	F	F	P	F	F	F	T	F	F	T
	01.02.01		Unlocking Sequence	T	T	ITL	–	M	–	T	F	P	F	P	F	P	F	F	F	F	T	*	T	T
	01.04.00		Fault State	T	T	?	–	–	–	P	F	P	F	P	F	P	F	P	F	F	F	F	F	T
	01.04.01		Fault Silenced State	T	T	?	–	–	M	A		A		A		P	F	F	F	T	T	F	F	T
	01.05.00		L/S Fault State Lock	F	F	?	–	–	–	D	F	D	F	D	F	P	F	D	F	F	F	F	F	T
	01.11.00		Overload Fault State	?	?	?	–	–	–	P	F	P	F	P	F	P	F	K	T	F	F	F	F	T

RESET PROCEDURE:

Motor Overload Fault State (Amber LED LD50 is illuminated and the Dok-Lok horn is slowly chirping.)

1. Press and hold Horn Silence Button (3-Button System)/Horn Silence #2 Button (5-Button System) until Horn Chirps (≈ 5 Seconds). Or
2. Press and Release Restraint O/L Button on Micro Controller Board.

When the motor overload has been reset, the Yellow LD50 LED will turn off and normal operation resumes. If Dok-Lok motor does not run after resetting the overload, check Motor Fuse 10FU1.

TROUBLESHOOTING

Setting Horn Override Code

To create a new or replace a forgotten code:

1. Press and hold DIAGNOSTIC button until horn chirps (≈ 5 seconds).

   or

   Without opening the control box cover, press and hold the #1 and #3 buttons on the cover until the horn chirps.
2. Enter the factory preset HORN OVERRIDE code: 1223. (horn will chirp)
3. Enter the new HORN OVERRIDE code. It can be 1 to 4 numbers in length.
4. Press the LOCK button. New HORN OVERRIDE code is enabled.

TROUBLESHOOTING

Diagnostics Mode

Diagnostic mode may be entered while restraint is in any state. To enter diagnostic mode:

1. Press and hold DIAGNOSTIC button until horn chirps (≈ 5 seconds).
2. Press LOCK button.
3. Press UNLOCK button.
4. Horn chirps and outside light is flashing RED. Controls are in first step of diagnostic mode.

   NOTE: Outside red light will remain flashing at all times except Step 10 (Diagnostic Sequence).
5. Start at Step 1 (Diagnostic Sequence).

If no buttons are pressed within a 5 minute period, controls will automatically exit to power up. To exit diagnostic mode at any time, press DIAGNOSTIC button.

Diagnostic Sequence

Press LOCK to advance, UNLOCK to reverse. C = Horn Chirp F = Off P = Pulsing / Flashing T = Steady On			OUTPUTS*																				
			BASE MICRO CONTROLLER BOARD																POWER BOARD				
			12VDC														RELAY		115/230VAC				
			Inside red light [ISR]	Inside red light [ISR]	Corner-Vu red light [CVU RD]	Corner-Vu green light [CVU GRN]	Leveler-Vu red light [LVU RD]	Leveler-Vu green light [L-VU GRN]	Pedestrian-Vu light [PVU LT]	Approach-Vu light [AVU LT]	Outside red light [OSR]	Outside green light [OSG]	Dok-lok horn [HORN]	MWL actuator [MWLACTR]	Outside alarm [MWL ALM]	Unidox ITC [UD ITC OUT]	Green light interlock [GLT ITL]	Green light interlock [GLT ITL]	K2 - security system interface or combined power unit [if equipped]	Motor ouptut #1 [M1/LOCK]	Motor ouptut #2 [M2/UNLOCK]	12VDC power supply ok
TERMINAL BLOCK NO.			J7.16	J7.17	J12.1	J12.2	J12.3	J12.4	J12.6	J12.5	J11.2	J11.1	J7.19	J15.2	J15.3	J15.4	J15.5	J9.3	J10.3	J5.4	J5.3	J2.1-6
POWER BOARD LEDS			–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	LD2	LD1	LD7
MICRO CONTROL BOARD LEDS			LD17	LD19	LD11	LD13	LD18	LD12	LD16	LD14	LD49	LD48	LD15	LD37	LD40	LD20	LD39	LD9	LD10	LD1	LD3	-
STEP	1	Diagnostics entered	F	F	F	F	F	F	F	F	P	F	C	F	F	F	F	F	F	F	F	T
	2	Check Inside red	T	F	F	F	F	F	F	F	P	F	F	F	F	F	F	F	F	F	F	T
	3	Check Inside green	F	T	F	F	F	F	F	F	P	F	F	F	F	F	F	F	F	F	F	T
	4	Check Corner-Vu red	F	F	T	F	F	F	F	F	P	F	F	F	F	F	F	F	F	F	F	T
	5	Check Corner-Vu green	F	F	F	T	F	F	F	F	P	F	F	F	F	F	F	F	F	F	F	T
	6	Check Leveler-Vu red	F	F	F	F	T	F	F	F	P	F	F	F	F	F	F	F	F	F	F	T
	7	Check Leveler-Vu green	F	F	F	F	F	T	F	F	P	F	F	F	F	F	F	F	F	F	F	T
	8	Check Pedestrian-Vu light	F	F	F	F	F	F	T	F	P	F	F	F	F	F	F	F	F	F	F	T
	9	Check Approach-Vu light	F	F	F	F	F	F	F	T	P	F	F	F	F	F	F	F	F	F	F	T
	10	Check Outside red light	F	F	F	F	F	F	F	F	T	F	F	F	F	F	F	F	F	F	F	T
	11	Check Outside green light	F	F	F	F	F	F	F	F	P	T	F	F	F	F	F	F	F	F	F	T
	12	Check Dok-Lok horn	F	F	F	F	F	F	F	F	P	F	T	F	T	F	F	F	F	F	F	T
	13	Check MWL actuator, Unidox outputs	F	F	F	F	F	F	F	F	P	F	F	T	F	T	F	F	F	F	F	T
	14	Check Green light interlock outputs	F	F	F	F	F	F	F	F	P	F	F	F	F	F	T	T	F	F	F	T
	15	Check K2 relay	F	F	F	F	F	F	F	F	P	F	F	F	F	F	F	F	T	F	F	T
	16	Horn chirps = end of sequence	F	F	F	F	F	F	F	F	P	F	C	F	F	F	F	F	F	F	F	T

***IF OUTPUT DOESN'T MATCH:**		
2-3, 12	Check	Control harness connection at chevron and micro controller boards
3-11	Check	Light bulb, wiring and terminal block connections
13-15	Check	Terminal block connections

Power supply LED and power supply fuse on power circuit board

TROUBLESHOOTING

Diagnostics Mode *Continued*

Steps 17-21 (Diagnostic Sequence) are used to test mechanical Inputs and Outputs of lock (motor, limit switches, etc.).

- When controls enter Steps 17-21, horn will chirp to alert operator to check hook position.
- If hook does not move to correct position, horn will double pulse and inside red light will be on.

Diagnostic Sequence (If Equipped)

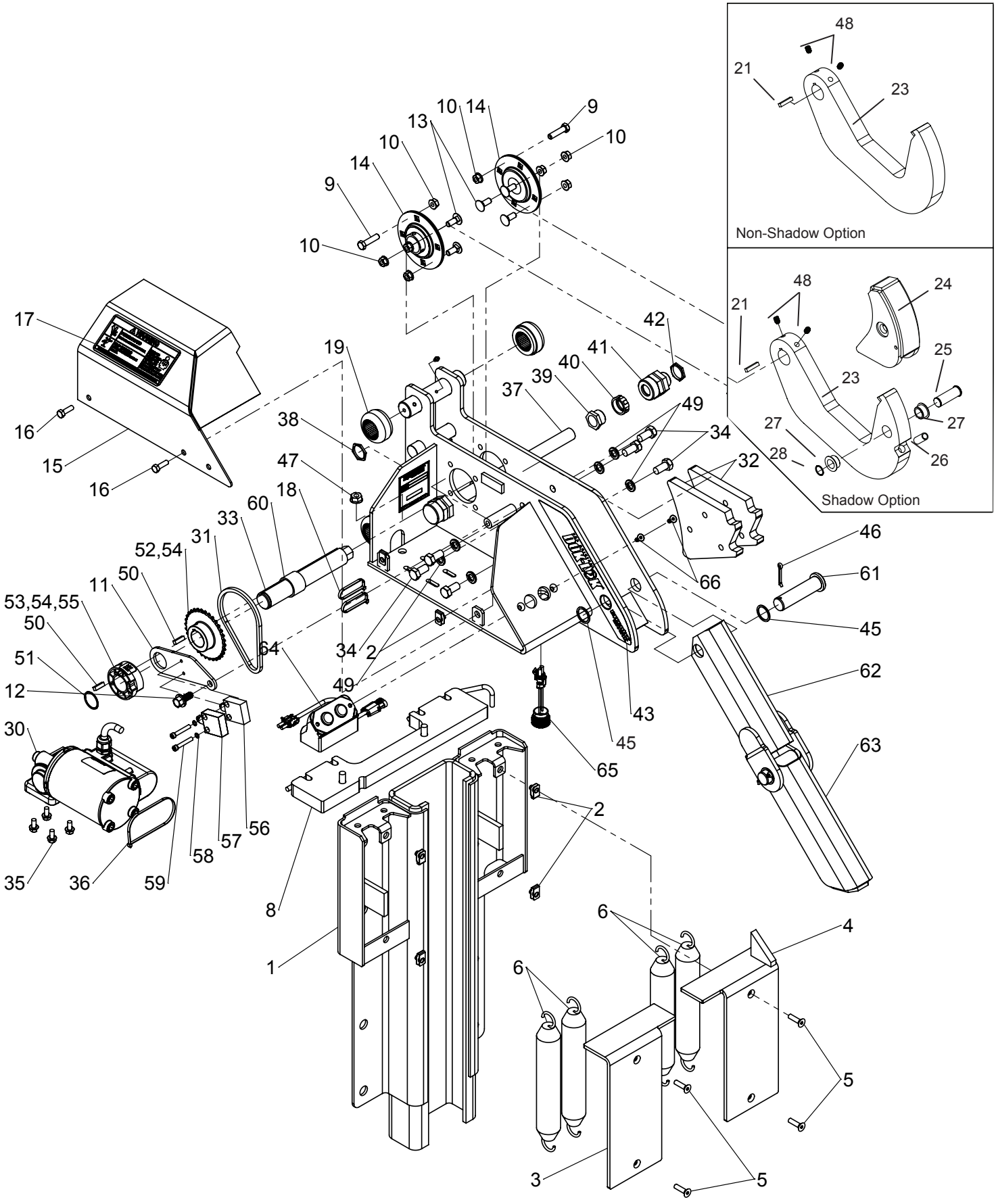
Press LOCK to advance, UNLOCK to reverse.

C = Horn Chirp
D = Double Pulse
F = Off
P = Pulsing / Flashing
T = Steady On

			MICRO CONTROL BOARD											POWER BOARD		
			INPUTS		OUTPUTS											
			FIELD		12VDC									115/230VAC		
			Dok-Lok limit switch 1 [SW1]	Dok-Lok limit switch 2 [SW2]	Inside red light [ISR]	Inside red light [ISR]	Corner-Vu red light [CVU RD]	Corner-Vu green light [CVU GRN]	Leveler-Vu red light [LVU RD]	Leveler-Vu green light [L-VU GRN]	Outside red light [OSR]	Outside green light [OSG]	Dok-lok horn [HORN]	Motor ouptut #1 [M1/LOCK]	Motor ouptut #2 [M2/UNLOCK]	12VDC power supply ok
TERMINAL BLOCK NO.			J13.1	J13.2	J7.16	J7.17	J12.1	J12.2	J12.3	J12.4	J11.2	J11.1	J7.19	J5.4	J5.3	J2.1-6
POWER BOARD LEDS			–	–	–	–	–	–	–	–	–	–	–	LD2	LD1	LD7
MICRO CONTROL BOARD LEDS			LD20	LD23	LD17	LD19	LD11	LD13	LD18	LD12	LD49	LD48	LD15	LD1	LD3	-
STEP	17	Horn chirps = entrance into motor sequence	?	?	T	F	T	F	T	F	P	F	C	F	F	T
	18a	Hook stores if not already stored	T	F	T	F	T	F	T	F	P	F	C	F	T	T
	18b	Hook runs to lower FAULT position	T	T	T	F	T	F	T	F	P	F	C	T	F	T
	19	Hook runs to locked position	F	T	T	F	T	F	T	F	P	F	C	T	F	T
	20	Hook runs to upper FAULT position	T	T	T	F	T	F	T	F	P	F	C	T	F	T
	21	Hook stores and system exits diagnostics	T	F	T	F	T	F	T	F	P	F	C	F	T	T
	FAULT	Diagnose general FAULT state	?	?	T	F	T	F	T	F	P	F	D	F	F	T
	IF OUTPUT DOESN'T MATCH:															
	FAULT	Check	Power supply LED & power supply fuse on power circuit board													
			Motor and verify wiring													
			Terminal block connections													
			SW1 and SW2 and verify wiring													
			Motor fuse on power circuit board													
			Voltage on M1 and M2 terminals to neutral													

PARTS

Carriage and Track 9in



PARTS

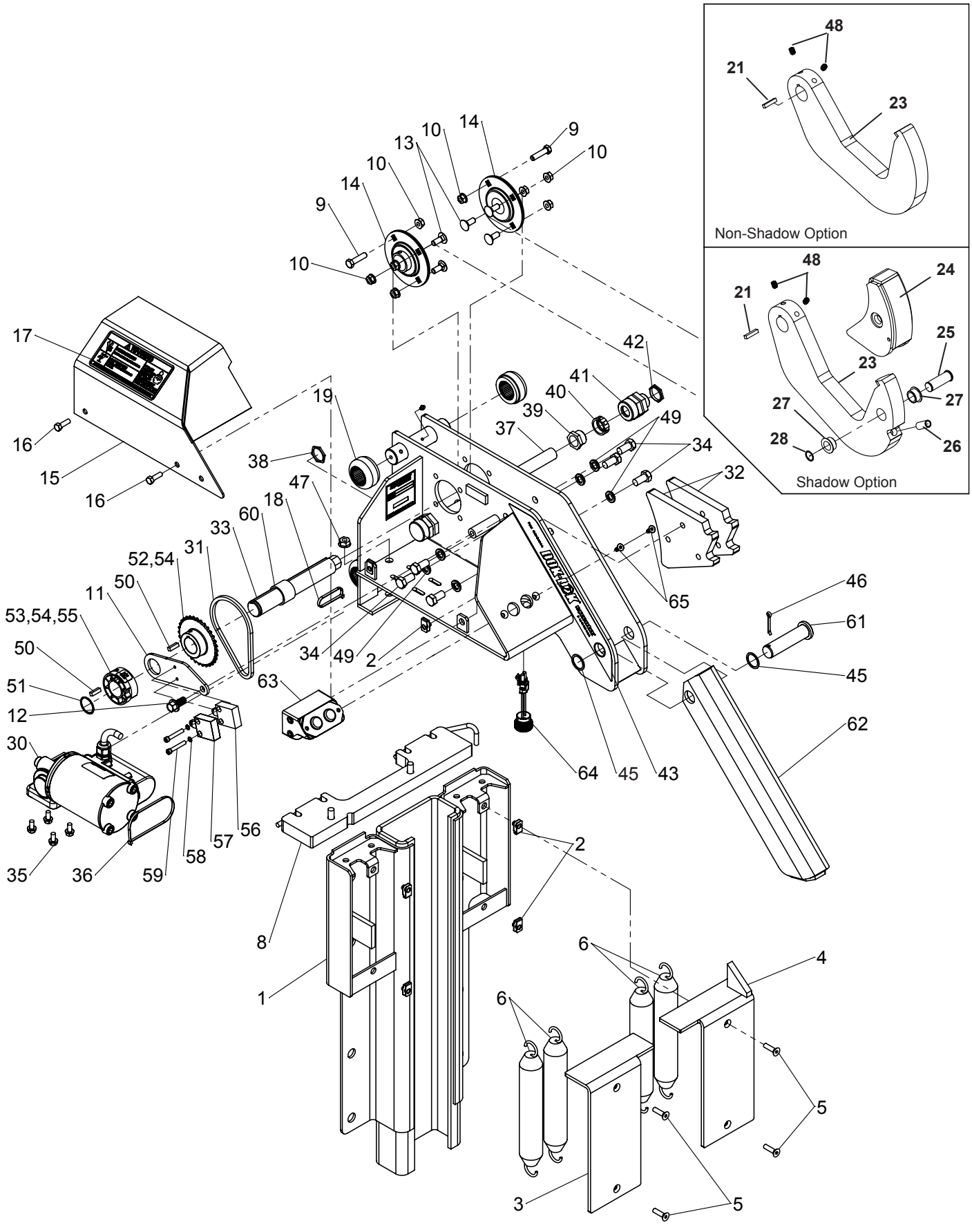
Carriage and Track 9in *Continued*

#	QTY	DESCRIPTION	PART #	
			SHADOW	NON-SHADOW
1	1	Roller Track	139912	139912
2	6	Cage Nut	56540	56540
3	1	Spring Cover, Left (facing dock)	139919	139919
4	1	Spring Cover, Right (facing dock)	139918	139918
5	4	Mounting Screw (5/16 - 18 x 1"L) flat head socket	18210	18210
6	4	Extension Spring	146443	146443
7*	15	Concrete Anchor, 5/8" dia. X 4"L (Powers #6942 or equivalent)	53150	53150
8	1	Spring Bar	139915	139915
9	2	Bolt, hook stop	68207	68207
10	8	Nut, serrated flange (3/8" - 16)	51564	51564
11	1	Switch Mount	140343	140343
12	1	Serrated Flange Bolt 7/16"-14 x 1" Long	140346	140346
13	6	Carriage Bolt, 3/8" - 16 x 1" Long	58022	58022
14	2	Bearing	06085	06085
15	1	Motor Cover	140044	140044
16	2	Bolt, 5/16" - 18 x 1" Long Hex	51627	51627
17	1	Decal, "Warning"	141990	141990
18	2	Wire Tie, .18 x 6.75" Long	55592	55592
19	4	Carriage Roller	15782	15782
20	2	Zerk Fitting	51169	51169
21	1	Hook Key (1/4" x 1/4" x 1-1/4" Long)	54818	54818
22	1	Barrier Assembly (includes items 25 - 29)	136784	N/A
23	1	Hook	136794	139761
24	1	Barrier Support	136785	N/A
25	1	Clevis Pin, 3/4" Dia x 2-1/8" Long, Stainless Steel	136797	N/A
26	1	Spring	137215	N/A
27	2	Flanged Bushing	136796	N/A
28	1	Retaining Ring Spiral	137218	N/A
29	1	Carriage	150714	150714
30	1	Motor	138158	138158
31	1	Chain	140063	140063
32	2	Hook Stop Plate	136775	136775
33	1	Hook Shaft	54987	54987
34	6	Hex Bolt 1/2"-13 x 1" Long	125347	125347
35	4	Bolt, 5/16"-18 x 5/8" Long	58069	58069
36	1	Wire Tie, 13 1/2" Long	55892	55892
37	1	Control Harness (item 38 is locknut for harness)	150709-01	150709-01
38	1	Lock Nut	55791	55791
39	1	Conduit Nipple	18204	18204
40	1	Conduit Bushing	57978	57978
41	1	Harness Cord Grip	66082	66082
42	1	Harness Lock Nut w/Gasket	55877	55877
43	1	Carriage Decal, Left (facing dock)	140207	140207
44	1	Carriage Decal, Right (facing dock)	140208	140208
45	2	Machinery Bushing, .06" Thk x 1.015"I.D.	19128	19128
46	2	Cotter Pin, .16"Dia x 1.5" Long	105865	105865
47	2	Nut, Serrated Flange (7/16"-14)	51549	51549
48	2	Set Screw, 3/8"-16 Knurled	139428	139428
49	6	1/2" Lockwasher	51831	51831
50	2	Key 1/4" x 1/4" x 7/8" Long	54864	54864
51	1	Retaining Ring, Spiral	56526	56526
52	1	Sprocket #35-32 Tooth	140340	140340
53	1	Cam	140344	140376
54	2	Set Screw, 5/16"-18 Knurled	56525	56525
55	1	Decal, Cam	18213	18213
56	1	LS1 Limit Switch	18211	18211
57	1	LS2 Limit Switch	18212	18212
58	2	Hi-Collar Lockwasher 1/4"	51802	51802
59	2	Socket Head Screw 1/4"-20 x 1 1/2" Long	51636	51636
60	1	Sprocket Spacer	147472	147472
61	1	Slope Extension Clevis Pin, 1"Dia x 3-1/2" Long	105495	105495
62	1	Slope Extension, Upper	140051	140051
63	1	Slope Extension, Lower	140052	140052
Approach-Vu				
64	1	Light Assembly Yellow/White	150665	150665
	1	Hole Plug, 2 Hole Black (W/O Approach-Vu)	150676	150676
	1	Hole Plug, 2 Hole Silver (W/O Approach-Vu)	150677	150677
*	1	Kit Replacement Approach-Vu Lens	152353	152353
65	1	Alarm Assembly	150687	150687
	1	Hole Plug (W/O Approach-Vu)	150678	150678
66	2	Screw 1/4-20 Flat Head Stainless	150717	150717

*Not Shown

PARTS

Carriage and Track 12in



PARTS

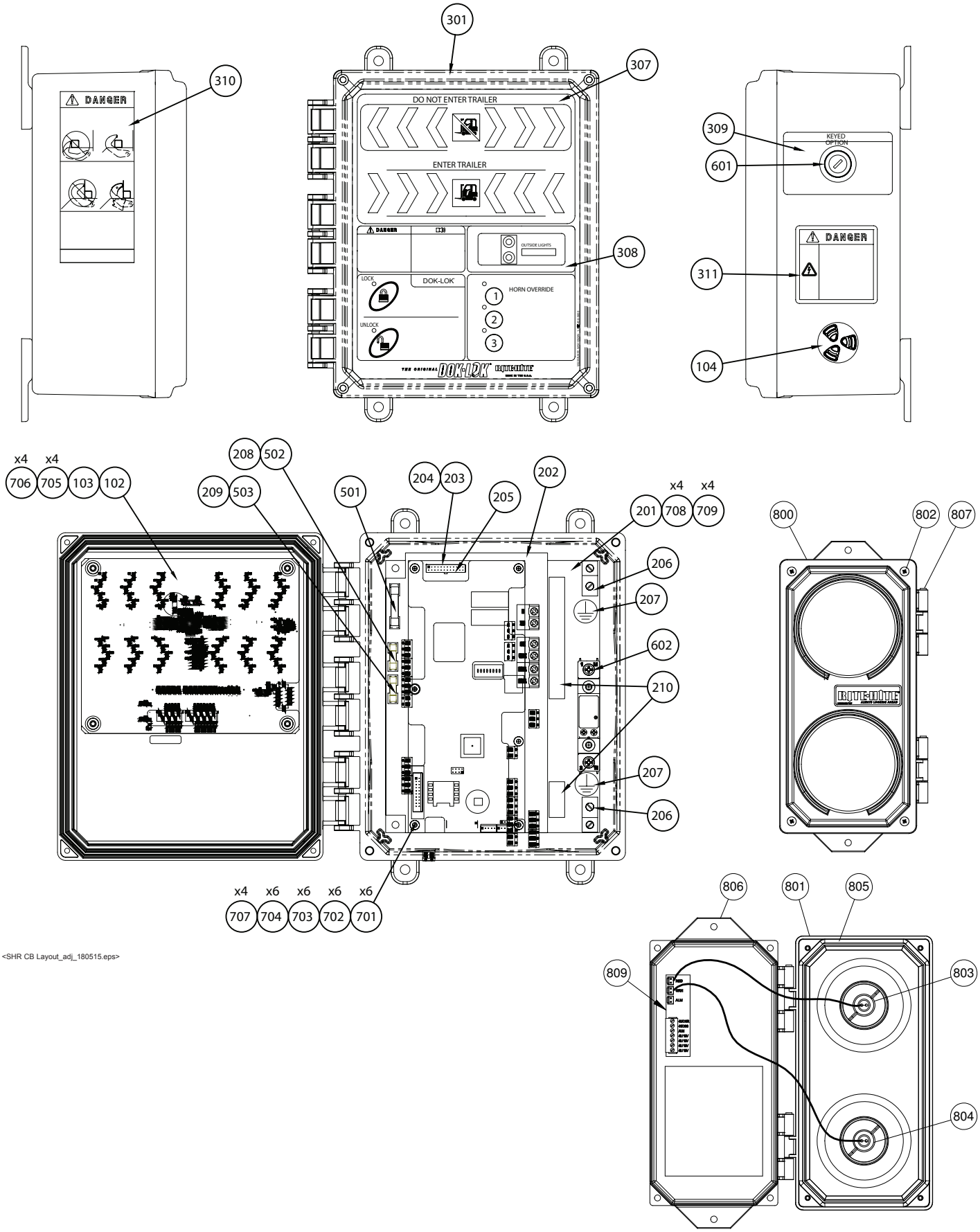
Carriage and Track 12in *Continued*

#	QTY	DESCRIPTION	PART #	
			SHADOW	NON-SHADOW
1	1	Roller Track	139912	139912
2	6	Cage Nut	56540	56540
3	1	Spring Cover, Left (facing dock)	139919	139919
4	1	Spring Cover, Right (facing dock)	139918	139918
5	4	Mounting Screw (5/16 - 18 x 1"L) flat head socket	18210	18210
6	4	Extension Spring	146443	146443
7*	15	Concrete Anchor, 5/8" dia. X 4"L (Powers #6942 or equivalent)	53150	53150
8	1	Spring Bar	139915	139915
9	2	Bolt, hook stop	68207	68207
10	8	Nut, serrated flange (3/8" - 16)	51564	51564
11	1	Switch Mount	140343	140343
12	1	Serrated Flange Bolt 7/16"-14 x 1" Long	140346	140346
13	6	Carriage Bolt, 3/8" - 16 x 1" Long	58022	58022
14	2	Bearing	06085	06085
15	1	Motor Cover	140044	140044
16	2	Bolt, 5/16" - 18 x 1" Long Hex	51627	51627
17	1	Decal, "Warning"	141990	141990
18	2	Wire Tie, .18 x 6.75" Long	55592	55592
19	4	Carriage Roller	15782	15782
20	2	Zerk Fitting	51169	51169
21	1	Hook Key (1/4" x 1/4" x 1-1/4" Long)	54818	N/A
22	1	Barrier Assembly (includes items 25 - 28)	136784	N/A
23	1	Hook	136794	139760
24	1	Barrier Support	136785	N/A
25	1	Clevis Pin, 3/4" Dia x 2-1/8" Long, Stainless Steel	136797	N/A
26	1	Spring	137215	N/A
27	2	Flanged Bushing	136796	N/A
28	1	Retaining Ring Spiral	137218	N/A
29	1	Carriage	150720	150720
30	1	Motor	138158	138158
31	1	Chain	140063	140063
32	2	Hook Stop Plate	136775	136775
33	1	Hook Shaft	54987	54987
34	6	Hex Bolt 1/2"-13 x 1" Long	125347	125347
35	4	Bolt, 5/16"-18 x 5/8" Long	58069	58069
36	1	Wire Tie, 13 1/2" Long	55892	55892
37	1	Control Harness (item 38 is locknut for harness)	150709-01	150709-01
38	1	Lock Nut	55791	55791
39	1	Conduit Nipple	18204	18204
40	1	Conduit Bushing	57978	57978
41	1	Harness Cord Grip	66082	66082
42	1	Harness Lock Nut w/Gasket	55877	55877
43	1	Carriage Decal, Left (facing dock)	138250	138250
44	1	Carriage Decal, Right (facing dock)	138249	138249
45	2	Machinery Bushing, .06" Thk x 1.015"I.D.	19128	19128
46	2	Cotter Pin, .16"Dia x 1.5" Long	105865	105865
47	2	Nut, Serrated Flange (7/16"-14)	51549	51549
48	2	Set Screw, 3/8"-16 Knurled	139428	139428
49	6	1/2" Lockwasher	51831	51831
50	2	Key 1/4" x 1/4" x 7/8" Long	54864	54864
51	1	Retaining Ring, Spiral	56526	56526
52	1	Sprocket #35-32 Tooth	140340	140340
53	1	Cam	140344	140376
54	2	Set Screw, 5/16-18 Knurled	56525	56525
55	1	Decal, Cam	18213	18213
56	1	LS1 Limit Switch	18211	18211
57	1	LS2 Limit Switch	18212	18212
58	2	Hi-Collar Lockwasher	51802	51802
59	2	Sock Head Screw 1/4-20 x 1 1/2" Long	51636	51636
60	1	Sprocket Spacer	147472	147472
61	1	Slope Extension Clevis Pin, 1"Dia x 3-1/2" Long	105495	105495
62	1	Slope Extension	140053	140053
Approach-Vu				
63	1	Light Assembly Yellow/White	150665	150665
	1	Hole Plug, 2 Hole Black (W/O Approach-Vu)	150676	150676
	1	Hole Plug, 2 Hole Silver (W/O Approach-Vu)	150677	150677
*	1	Kit Replacement Approach-Vu Lens	152353	152353
64	1	Alarm Assembly	150687	150687
	1	Hole Plug (W/O Approach-Vu)	150678	150678
65	2	Screw 1/4-20 Flat Head Stainless	150717	150717

*Not Shown

PARTS

Electrical



<SHR CB Layout_adj_180515.eps>

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PARTS

Electrical *Continued*

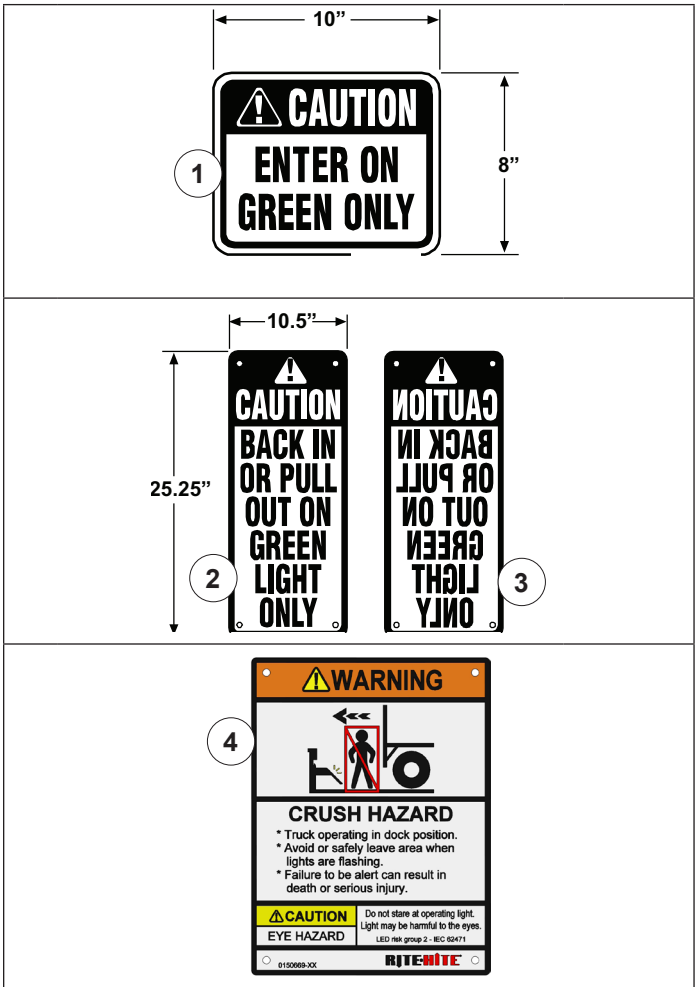
#	QTY	DESCRIPTION	PART #
1	1	Control box assembly, Complete	863.xxx**
102	1	LED chevron/outside light monitor circuit board	141579
103	1	LED chevron/outside light monitor circuit board cover	144835
104	1	Horn, 12VDC, N4X	141573
201	1	Pre-drilled control box sub panel	144757
202	1	Power circuit board, Blank	141580
	1	Power circuit board, Including Fuses (115 Volt Power)	142126-04
	1	Power circuit board, Including Fuses (230 Volt Power)	142126-05
203	1	Standard Micro Control Board, Less Advanced Controls	141578-861D100
	1	Standard Micro Control Board, Including Advanced Controls	141578-861D102
	1	Micro Control Board with Program Options	141578-xxx**
204	1	Micro circuit board cover	144802
205	1	Control wiring harness	144864-01
206	2	Ground lug	55902
207	2	Ground lug decal	105454
208	1	Fuse cover, 5mm x 20mm	144544
209	1	Fuse cover, 5mm x 20mm	144544
210	1	Panel decals	146324
301	1	Standard Enclosure with decals, English, Less Advanced Controls	865.100
	1	Standard Enclosure with decals, English, Including Advanced Controls	865.102
	1	Enclosure with decals other than above	865.xxx**
302*	1	Electrical schematic decal	870.xxx**
303*	1	Fuse replacement decal	144849
304*	1	Full load amperage and voltage/phase decal	872.xxx**
305*	1	Program configuration decal	862.xxx**
307	1	Cover decal with membrane push buttons	141499
308	1	Dok-Lok inlay decal, 5 button coded horn override (Advanced Control Package)	141502
	1	Dok-Lok inlay decal, 3button horn override (Less Advanced Control Package)	144803
309	1	Horn override enable decal (If equipped)	144831
	1	Control power ON/OFF decal (If equipped)	144829
	1	Unlock enable decal (If equipped)	144830
310	1	Danger/warning decal	144774
311	1	Lockout/tagout/multiple disconnect warning decal	114331
313*	1	Circuit board UL rating decal	144859
501	1	6A fuse, 0.25" x 1.25" Motor Fuse (115V control box)	65263
502	1	1A fuse, 5mm x 20mm Solenoid Fuse (115V control box)	142092
503	1	1.5A fuse, 5mm x 20mm Power Supply Fuse	150984

#	QTY	DESCRIPTION	PART #
601	1	Selector switch, keyed (If equipped)	125194
602	1	Solid state relay, 12VDC, 75A, with hardware (If equipped)	144877
701	6	Standoff, Male/Female, #6-32, 0.75" length	142100
702	6	Standoff, Male/Female, #6-32, 1.625" length	142101
703	6	Screw, round head, #6-32, 0.25" length	133366
704	6	Lock washer, #6	51826
705	4	Rubber grommet	142102
706	4	Screw, black trim washer, #8-32, 0.5" length	146031
707	4	Standoff, Male/Female, #6-32, 0.875" length	115871
708	4	Screw, Round Head, #10-32, 0.5" length	51629
709	4	Lock Washer, #10	51762
800	1	LED Outside Light Assembly RED/GRN, Complete	155402
801	1	LED Outside Light Assembly RED/GRN, Doesn't Include Base	155403
802	4	SCR #10-32X.438 Mach Rnd Captive Blk	155416
803	1	LED Outside Light Bulb, Red	155406
804	1	LED Outside Light Bulb, Green	155405
805	1	Rubber Gasket	155399
806	1	Outside Light Enclosure Base	155400
807	2	Hinge Pin	155398
809	1	Circuit Board	155413

*Not Shown
**Consult Rite-Hite for part number.

PARTS

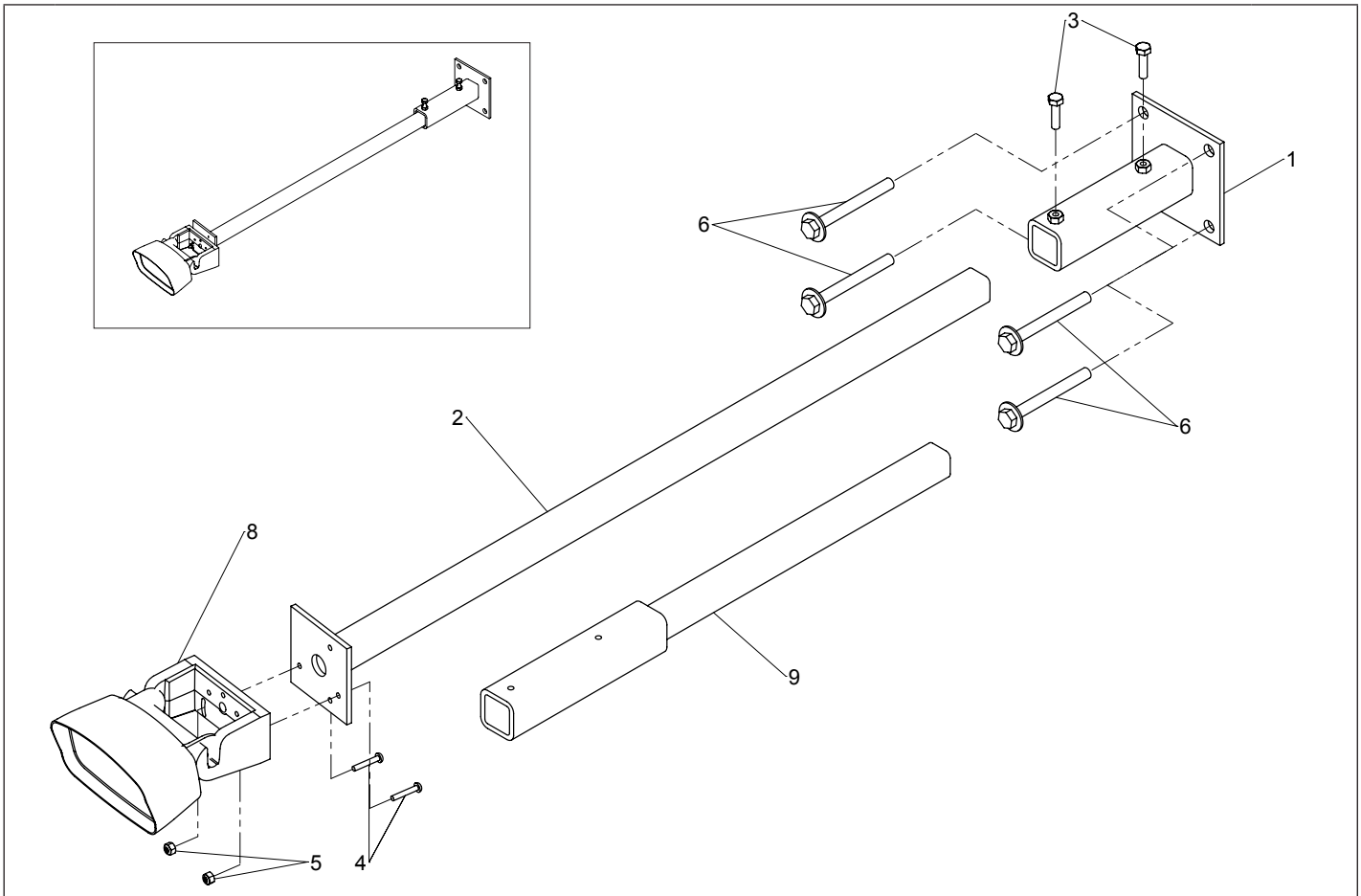
Signs



#	QTY	DESCRIPTION	PART #
1	1	Interior Warning Sign	56095
2	1	Narrow Sign (normal letters)	56112
3	1	Narrow Sign (mirror letters)	56113
4	1	Approach-Vu Sign	150669

PARTS

Approach-Vu™ Truck Sensor



#	QTY	DESCRIPTION	PART #
1	1	Mounting Weldment - Base	150695
2	1	Mounting Weldment - Extension	150691
3	2	Bolt .25-20 x 1"	136880
4	2	Screw #8-32X1.00 Phillips Head, ZP	51689
5	2	Nut #8-32 Nylock, ZP	141611
6	4	Anchor Concrete .375 dia X 3.5L Hex Head, ZP	105046
7*	1	Installation Instructions, Truck Sensor	150697
8	1	Truck Sensor	150688
9	1	Extension Weldment (for shelters over 36" deep)	151306

*Not Shown

Rite-Hite Company, LLC and its affiliates (collectively "Rite-Hite") warrant that the Product sold to the Owner will be free of defects in design, materials and workmanship (ordinary wear and tear excepted) for the periods set forth below ("Limited Warranty").

Five (5) year parts and Two (2) years labor warranty with purchase of Advanced Controls Package.
Five (5) year parts and Five (5) years labor warranty with purchase of Advanced Controls Package and a signed P.M.P. (Preventative Maintenance Program) with an Authorized Rite-Hite representative.
Two (2) year parts and One (1) year labor warranty with Base Controls Package.

REMEDIES
Parts: Rite-Hite's obligations under this Limited Warranty are limited to repairing or replacing, at Rite-Hite's option, any part which is determined by Rite-Hite to be defective during the applicable warranty period. Such repair or replacement shall be Rite-Hite's sole obligation and the Owner's exclusive remedy under this Limited Warranty.
Labor: Rite-Hite will provide warranty service without charge for labor per the specified warranty period. Thereafter, a charge will apply to any repair or replacement under this Limited Warranty.

CLAIMS Claims under this Limited Warranty must be made (i) within 30 (thirty) days after discovery and (ii) prior to expiration of the applicable warranty period. Warranty period commences on the date of shipment. Claims shall be made in writing or by contacting the representative from whom the Product was purchased directly. Owner must allow Rite-Hite or its agent, a reasonable opportunity to inspect any Product claimed to be defective and shall, at Rite-Hite's option, either (x) grant Rite-Hite or its agent access to Owner's premises for the purpose of repairing or replacing the Product or (y) return of the Product to the Rite-Hite, F.O.B. Rite-Hite's factory.

NOT WARRANTED Rite-Hite does not warrant against and is not responsible for wear items such as fuses, batteries, bulbs, vision and seals. No implied warranty shall be deemed to cover damages that result directly or indirectly from: (i) the unauthorized modification or repair of the Product, (ii) damage due to misuse, neglect, accident, failure to provide necessary maintenance, or normal wear and tear of the Product, (iii) failure to follow Rite-Hite's instructions for installation, failure to operate the Product within the Product's rated capacities and/or specified design parameters, or failure to properly maintain the Product, (iv) use of the Product in a manner that is inconsistent with Rite-Hite's guidelines or local building codes, (v) movement, settling, distortion, or collapse of the ground, or of improvements to which the Products are affixed, (vi) fire, flood, earthquake, elements of nature or acts of God, riots, civil disorder, war, or any other cause beyond the reasonable control of Rite-Hite, (vii) improper handling, storage, abuse, or neglect of the Product by Owner or by any third party.

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170714

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